Fate Contagion in the Juggler Map and the Almost-All Reduction of Termination

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Abstract

The Juggler map is the integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

Every orbit has exactly one of three fates: it reaches 1, it enters a nontrivial cycle, or it is unbounded. It is conjectured that the first fate is the only one. This paper does not prove that conjecture. Its main theorem is structural: every nonempty set A of positive integers that is closed under taking preimages ($J(n) \in A \Rightarrow n \in A$) — in particular every fate class that occurs at all — satisfies $\sum_{n \in A,\, n \le x} 1/n \ge c\,(\log x)^{\lambda}$ for every $\lambda < \lambda^{**} = 0.4480\ldots$, whereas $\sum_{n \le x} 1/n \sim \log x$. Fates are contagious. The mechanism is that the one-step preimages of the Juggler map are large and structured: the even preimages of m fill the interval $[m^2, (m+1)^2)$, and the two-step preimages through an even middle state fill a positive proportion of the fiber $\lfloor n^{3/4} \rfloor = m$, on which the parity of $\lfloor n^{3/2} \rfloor$ sweeps. The proof is elementary at this depth; it needs positive proportions on fibers, not equidistribution.

Three consequences follow. First, a failure set is generated by its odd members, which are the odd preimages of a sparse set of odd images (odd generation). Second, the Juggler conjecture is equivalent to a Tao-type almost-all statement with a *bounded* target and a logarithmic rate: every positive integer reaches 1 if and only if all but $O(y(\log y)^{-e})$ odd starts in $(y, 2y]$ enter the certified interval $[1, N_0]$, for some $e > 1 - \lambda^{**}$. For the Collatz map no such equivalence is available, because the known lower bounds for Collatz preimage trees are of polynomial size, which a logarithmic error rate cannot see. Third, that almost-all statement follows from parity control on itinerary cylinders of depth $C \log_2 \log y$, in a hierarchy of forms whose weakest is a single exponential moment of the odd count on live starts; we determine what that weakest form does and does not need, show that no bounded-depth cylinder statement implies it, and quantify the depth uniformity an analytic cylinder bound would have to have. An exact first-letter decomposition of the failure set has a single free term, and that term is the infinite-depth live mass controlled by the same hypothesis: the termination problem has one frontier statement. The We also determine the ceiling of the contagion recursion itself: with ideal fiber shares its exponent solves $2^{-\lambda} + \frac{1}{3}(\frac{3}{2})^{\lambda} = 1$ and equals 1, attained on the fair-coin backward path, so the recursion is capped not by its own structure but by two named obstructions — a fiber-length threshold and the same parked kernel that blocks the analytic route. The exact combinatorial layer is formalized in Lean 4; the counting is a human proof; numerical experiments are reported as observations only. No fate is excluded.

1. Introduction

The Juggler sequence was introduced by Pickover [1] and is sequence A094683 of the OEIS [2]. Starting from any positive integer, the orbit under J appears always to reach 1; this has been verified for all starts up to $3.5 \cdot 10^8$ [11]. The map belongs to the family of floor-power maps, and its dynamics resembles the $3x + 1$ problem [3, 4] in one respect and differs from it in another. The resemblance: a parity decides whether the orbit goes up or down, and the heuristic model of the orbit is a random walk with negative drift, here on the scale $\log\log n$ with steps $+\log_2 3 - 1$ (odd) and -1 (even). The difference: the Juggler map moves by *powers*, so a single contracting certificate sends n to n^e with $e < 1$, and the one-step preimages of an integer are not a residue class but an interval.

This paper is about the second feature and what it forces. The narrative has four steps.

Contagion. Let R be the set of starts that reach 1, F its complement, $B(C)$ the basin of a nontrivial cycle C, and D the set of divergent starts. Each is *backward-closed*: if $J(n)$ belongs to it, so does n. Theorem 1 proves that every nonempty backward-closed set is large in logarithmic density, with an explicit exponent, by a downward recursion over two exact productions. Applied to the fate classes: if a single start fails to reach 1, the failures have logarithmic count $\gg (\log x)^{0.448}$ and, on infinitely many dyadic blocks, natural density $\gg (\log y)^{-0.55}$. This excludes no fate; it fixes the quantitative shape of the trichotomy.

Odd generation. A set that is closed both forwards and backwards and avoids 1 is the E-forest (iterated even preimages) over its odd members, and its odd members are the odd preimages of its intersection with the sparse set $S = \{\lfloor m^{3/2} \rfloor :\ m \text{ odd}\}$ (Theorem 2). An arbitrary failure set is thereby replaced by sparse odd-image seeds together with deterministic even ancestry. This is the geometry that the later sections use.

The almost-all equivalence. Tao [6] proved that almost all Collatz orbits (in logarithmic density) attain almost bounded values, with a target $f(N) \to \infty$ arbitrarily slowly. For the Juggler map, contagion and odd generation together turn a bounded-target statement into the conjecture: every positive integer reaches 1 if and only if all but $O(y(\log y)^{-e})$ odd starts in $(y, 2y]$ enter $[1, N_0]$, for some $e > 1 - \lambda^{**} = 0.552$ (Theorem 3). The threshold is the complement of the contagion exponent. For Collatz the analogous implication is not available: the known lower bound for a preimage tree below x is of polynomial size, $x^{0.84}$ (Krasikov–Lagarias [7]), and a set of that size is invisible to a logarithmic error rate.

The frontier. What does the almost-all statement itself need? Because descent is by powers, the exponent walk of an itinerary has to travel only $L(y) = \log_2(\log 2y / \log N_0)$ units to bring a start of size y below N_0, and a Chernoff count of the itineraries whose walk fails to do so within $C\,L(y)$ steps gives the rate, provided the itinerary cylinders of that depth are not over-populated (Theorem 4). The depth is $O(\log\log y)$, where cylinders are large; for the Collatz map the corresponding depth is $\Theta(\log y)$, where they have single elements. The hypothesis has a hierarchy of forms; the weakest is one exponential moment of the odd count on the starts still above the floor, or equivalently a no-momentum statement: odd-heavy prefixes do not bias the next parity toward odd, on average over depths. This form is insensitive to any $o(\log\log y)$ initial depths and to the all-odd tower cylinders, no bounded-depth cylinder statement implies it, and an analytic cylinder bound can feed it only if its saving exponent decays slower than $2^{-d/C}$ in the depth d. Finally, an exact first-letter decomposition of the failure set has a single free term, and that term is the infinite-depth live mass of the OO cylinder — the same quantity (Theorem 5). The termination problem therefore has one frontier statement, stated in Section 12.

1.1 Main results

Write $\ell_A(u, v] = \sum_{n \in A,\, u < n \le v} 1/n$. Throughout, N_0 is any integer with $[1, N_0] \subseteq R$ (the Lean-verified value is 260; the certified computational value is $3.5 \cdot 10^8$ [11]).

Theorem 1 (fate contagion). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $\lambda < \lambda^{**} = 0.4480\ldots$, the root of $2^{-\lambda} + \frac{1}{9}(\frac{3}{8})^\lambda + \frac{2}{9}(\frac{3}{4})^\lambda = 1$. There are $c > 0$ and x_0, depending on A and λ, with

$$\sum_{\substack{n \in A \\ n \le x}} \frac{1}{n} \ \ge\ c\,(\log x)^\lambda \qquad (x \ge x_0).$$

Consequently each fate class realized by at least one start — R, the basin of any existing nontrivial cycle, the divergent set — satisfies this bound, and on infinitely many dyadic blocks has natural density $\gg (\log y)^{\lambda - 1}$. (Theorem 5.3, Corollaries 5.4, 5.5.)

The exponent 0.4480 is not the method's ceiling. Section 5.7 determines that ceiling exactly: writing $\rho_w = 2^{-a}(3/2)^b$ for the source scale of a production word with a letters E and b letters O, the ideal coefficient is $2^{-|w|}/\rho_w$ — a formula that returns every coefficient used here — and if every O-production realized the fair fiber share then λ would solve $2^{-\lambda} + \frac{1}{3}(\frac{3}{2})^{\lambda} = 1$, whose root is $\lambda = 1$, attained on the fair-coin backward path (Proposition 5.12). Two obstructions stop the list short: the ideal share is available only for $\rho_w \leq \frac{1}{2}$, because the fiber has length $P^{1-\rho_w}$ (Proposition 5.13), and a run of r consecutive O's needs Paper B at depth $r+1$, which the parked kernel K_3 blocks at $r = 4$. The last rung before that wall is $\lambda \leq 0.8414$. The $r = 1$ words — those with no two consecutive O's — cost no Paper B estimate at all, because an isolated O is absorbed by the E after it into an exact $\lfloor \cdot^{3/4} \rfloor$; the first of them, $OEOEE$, would raise 0.4480 to 0.4801 unconditionally and 0.5392 to 0.5665, and the family $(OE)^{k-1}OEE$ telescopes exactly onto the ideal depth-two recursion, closing the gap to 0.4927.

Theorem 2 (odd generation). Let A be forward- and backward-closed with $1 \notin A$. Every $n \in A$ descends by even steps to an odd member $m \geq 3$ of A; for odd n, $n \in A \iff \lfloor n^{3/2} \rfloor \in A$; and $A \neq \emptyset$ iff A contains the image $\lfloor m^{3/2} \rfloor$ of some odd $m \geq 3$. In particular every positive integer reaches 1 iff no odd image fails to. (Theorem 6.1; Lean.)

Theorem 3 (the conjecture as an almost-all statement). The following are equivalent: (i) every positive integer reaches 1; (ii) for some $\lambda < \lambda^{**}$, the starts $n \leq x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^{\lambda})$; (iii) for some $e > 1 - \lambda^{**} = 0.5520\ldots$ and all large y, $\#\{n \text{ odd} \in (y, 2y] : n \notin R\} \leq y(\log y)^{-e}$. (Corollary 7.1, Theorems 7.2, 7.3.)

Theorem 4 (the frontier reduction). Let $L(y) = \log_2(\log 2y/\log N_0)$, $d(y) = \lceil CL(y) \rceil$, $p_C = (1 - 1/C)/\log_2 3$ and $e(C) = C\, D(p_C \| \frac{1}{2})/\ln 2$. Each of the following hypotheses implies (iii) of Theorem 3 with $e = e(C) - \varepsilon$, hence the conjecture for $C \geq 20$ (unconditionally; $C \geq 18$ under the conditional Appendix C):

(a) *cylinder form* $\mathrm{H}(C, A)$: no O-rooted itinerary cylinder of depth $d(y)$ exceeds its fair share $2^{-(d-1)}y/2$ among odd starts by more than $y(\log y)^{-A}$, $A > C + e(C)$ (Theorem 8.3);

(b) *one-sided form* $\mathrm{H}_q(C, A)$: every cylinder of depth $1 \leq t < d(y)$ sends at most a fraction $q < \log 2/\log 3$ of its members, plus $y(\log y)^{-A}$, to an odd next state (Theorem 9.1; $C \geq 44$ at $q = 0.55$);

(c) *pressure form* $\mathrm{P}_\theta(C)$: $\frac{1}{N}\sum_{n \text{ odd},\ \tau(n)>d} e^{\theta o_d(n)} \leq (\frac{1}{2}(1 + e^{\theta}))^d e^{o(d)}$ at $\theta = \log(p_C/(1 - p_C))$, where τ is the entrance time into $[1, N_0]$ and o_d the odd count of the first d letters; and its *no-momentum form* $\sum_{t<d}(s_\theta(t) - q)^+ = o(d)$ for the tilted odd share s_θ of live starts (Theorem 9.2, Proposition 9.3).

None of these hypotheses is proved here. Form (c) is insensitive to any $o(\log\log y)$ initial depths and tolerates tower cylinders biased up to odd share 0.836; no cylinder statement of bounded depth implies any of (a)–(c) (Section 9.3).

Theorem 5 (the exact decomposition and the single frontier). The logarithmic density φ_A of a two-way closed set satisfies the first-letter identity (6.1), whose single free term ψ_A is the share of A among OO-type odd starts. For $A = F$ that term is the infinite-depth live mass of the depth-two cylinder OO, and is bounded by every hypothesis of Theorem 4 (Proposition 10.1). The two halves of (6.1) are the contagion recursion of Theorem 1 and an upper recursion for the failure density, with the same critical exponent $1 - \lambda_{\text{ideal}} = 0.5073$ as the rate threshold of Theorem 3 up to the fiber-constant gap; the upper recursion alone never yields $F = \emptyset$ (Corollary 10.2).

Proposition (depth-uniformity budget). A cylinder-count route to Theorem 4(a) whose only error term at depth d is $O(y^{1-\delta_d})$ with $\delta_d = \delta_0 2^{-cd}$ reaches the depth $C \log_2 \log y$ iff $cC < 1$ (Proposition 10.3). Weyl differencing has $c \geq 1$.

Under a localization of the triple parity discrepancy of nested floor powers to sub-dyadic intervals — stated as an explicit hypothesis in Appendix C; it is Theorem 4.12 of the working draft [12], and its status is described there — the exponent of Theorem 1 improves to $\lambda^{***} = 0.5392\ldots$, the rate threshold of Theorem 3 to 0.4608, and the least depth constant of Theorem 4 to $C \geq 18$. Nothing in Sections 2–12 depends on Appendix C.

1.2 The three fates as currently constrained

The three fate classes are the three Moirai: Atropos cuts the thread (the class R), Lachesis measures an allotted length (a cycle basin), Clotho spins without end (divergence). What is known about each, all from outside this paper:

- *Atropos.* Every $n \le N_0 = 3.5 \cdot 10^8$ reaches 1 (independently certified computation, [11, Section 5]); the Lean-verified floor is $N_0 = 260$. Paper B [12] proves that 7/8 of all starts descend below themselves within five steps, with a power saving on dyadic blocks.
- *Lachesis.* A nontrivial cycle has minimum above $3.5 \cdot 10^8$ and period at least 780239 [11]; it has at least four even steps; its odd share is $\log 2/\log 3$ up to $O(\Lambda/L)$, the critical share at which the exponent walk has zero drift.
- *Clotho.* A divergent orbit has unbounded exponent walk; its record jumps are quantized to the $\log_2 3$ lattice [11]. Nothing excludes it.

Theorem 1 adds one sentence to each: whichever of the three acts at all acts on a set of logarithmic count $\gg (\log x)^{0.448}$.

1.3 Related work

Collatz parity vectors. Terras [4] and Everett [5] showed that the first k parities of the Collatz orbit of n are determined by $n \bmod 2^k$ and equidistributed exactly, and deduced that almost all starts have finite stopping time (descend below themselves). The analogous exact statement fails for the Juggler map: the parities of $\lfloor n^{3/2} \rfloor$, $\lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor$, ... are Diophantine, not algebraic, and their equidistribution is proved only to depth four (Paper B, [12]).

Tao's theorem. Tao [6] proved that for any $f(N) \to \infty$, almost all Collatz orbits (in logarithmic density) attain a value below $f(N)$. The bounded-target version is not available, and the renewal machinery through the Syracuse random variables is the heart of the proof. Section 7.2 explains, as a structural analogy and not as a transfer of methods, which parts of that picture change for the Juggler map.

Preimage trees. Krasikov and Lagarias [7] proved that the Collatz preimage tree of any integer has $\ge x^{0.84}$ elements below x. The relevant point for this paper is the scale: a set of polynomial size $x^{0.84}$ has logarithmic count that may tend to zero, so a bound of the form $y(\log y)^{-e}$ on failures says nothing about it. For the Juggler map the even preimages of m are an interval of length $2m+1$, and Theorem 1 gives every preimage tree logarithmic count $\gg (\log x)^{\lambda}$.

Floor-power maps and exponential sums. The parity of $\lfloor n^{3/2} \rfloor$ along an arithmetic progression is a question about $\{n^{3/2}/2\}$; Section 4 uses only an elementary sweep argument and, on even blocks, the second-derivative test with Vaaler's approximation [10], as for Piatetski-Shapiro sequences. Deeper nesting is Paper B's subject.

Companion papers. Paper A [11] proves period lower bounds for a hypothetical cycle from a certified descent floor (cycle financing, walk-charge envelope); Paper B [12] proves parity equidistribution of nested floor powers to depth four with power savings. Neither is reproved here; both enter as citations, and one statement of the type Paper B proves enters Appendix C as an explicit hypothesis.

1.4 Verification

Statements carry one of four roles. *Lean*: the exact combinatorial layer is formalized in Lean 4 (`formal/Problems/Juggler/FateContagion.lean`, no `sorry`; names in Appendix A). *Human proof*: the analytic and probabilistic counting. *Verified computation*: exact integer computations (the descent floor is Paper A's; the closure of $[1, 260]$ under the two productions up to 10^9 is computed here). *Observation*: numerical experiments that check hypotheses and constants; they prove nothing and are labelled wherever they appear.

Result	Status
Fate classes closed; trichotomy; exclusion (Lemma 2.1)	Lean
Even block is an interval (Lemma 3.1)	Lean
OE fiber is an interval; cell identity (Lemma 3.2)	Lean
Odd generation (Theorem 6.1)	Lean
Envelope descent into the floor (Lemma 8.1)	Lean, on Paper A's power envelope
Sweep lemma, monotone pairing, fiber parity, thin fibers (Lemmas 4.1, 4.1', 4.2–4.3)	human proof
Block average (Proposition 4.4), $C_0 = 250$ explicit	human proof
Recursion lemma and contagion (Lemma 5.1, Theorem 5.3)	human proof
First-letter identity (6.1)	human proof (exact combinatorics)
Almost-all equivalence (Theorems 7.2, 7.3)	human proof
Chernoff, Azuma, exponential-moment arguments (Sections 8–10)	human proof
Localized triple discrepancy (Appendix C)	hypothesis, conditional
Numerical experiments (Section 11)	observation

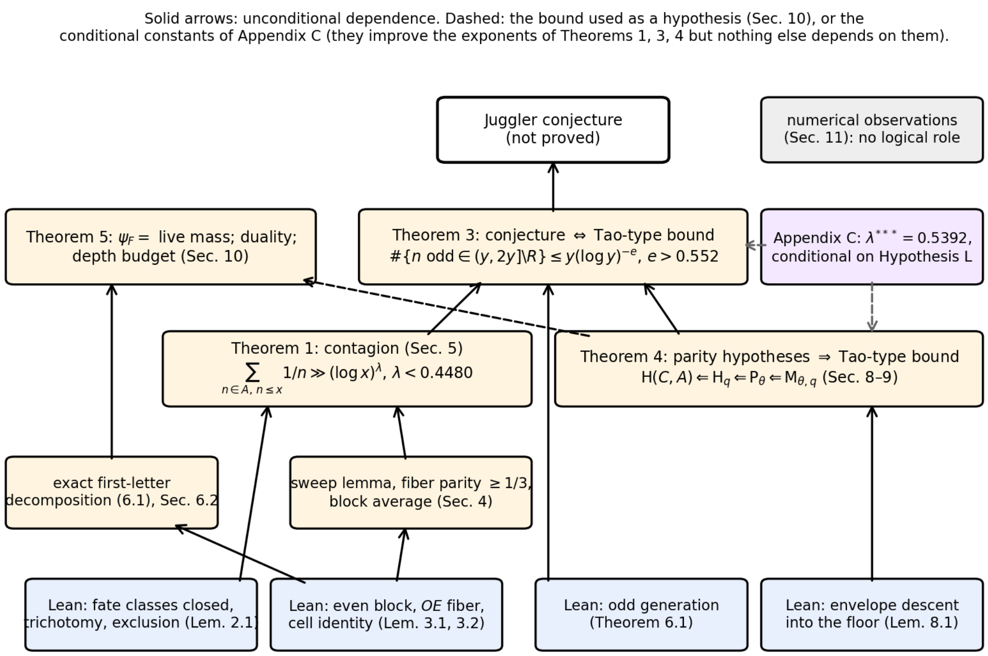

Figure 1: Logical dependencies. Solid arrows are unconditional; dashed arrows mark the Tao-type bound used as a hypothesis in Section 10 and the conditional constants of Appendix C.

2. Fate classes and closure

$\mathbb{N} = \{1, 2, \dots\}$. For $A \subseteq \mathbb{N}$ and $0 \le u < v$ write

$$\ell_A(u, v] = \sum_{\substack{n \in A \\ u < n \le v}} \frac{1}{n}, \qquad g_A(t) = \ell_A(e^{t/2}, e^t].$$

For $A = \mathbb{N}$, $g(t) = t/2 + O(e^{-t/2})$.

Definition. A is *backward-closed* if $J(n) \in A$ implies $n \in A$, and *forward-closed* if $n \in A$ implies $J(n) \in A$.

Lemma 2.1 (fate classes). Let $R = \{n : \exists k,\ J^k(n) = 1\}$, $F = \mathbb{N} \setminus R$, $B(m) = \{n : \exists k,\ J^k(n) = m\}$, and $D = \{n : \forall B\ \exists k,\ J^k(n) > B\}$. Each of $R, F, B(m), D$ is backward-closed; R, F, D are also forward-closed. Every $n \geq 1$ lies in R, or in $B(m)$ for some $m \geq 2$ on a cycle, or in D, and these possibilities exclude one another.

Proof. Backward closure is the definition of each set (for D: $J^k(J(n)) = J^{k+1}(n)$). Forward closure of R: if $J^k(n) = 1$ then $J^{k-1}(J(n)) = 1$ for $k \geq 1$, and $J(1) = 1$. Forward closure of D: given B, pick k with $J^k(n) > \max(B, n)$; then $k \geq 1$ and $J^{k-1}(J(n)) > B$. Trichotomy: a bounded orbit repeats, a repeat is a cycle, a cycle through 1 is the fate R. Exclusion: an orbit that reaches 1 or enters a cycle is bounded; an orbit that reaches 1 and passes through a cycle state $m \geq 2$ of period L would force $m = J^{qL}(m) = 1$. Lean: `reachesOne_backwardClosed`, `not_reachesOne_backwardClosed`, `ancestor_backwardClosed`, `escapes_backwardClosed`, `reachesOne_floorPower`, `escapes_floorPower`, `fate_trichotomy`, `reachesOne_not_escapes`, `cycle_basin_not_escapes`, `reachesOne_not_cycle_basin`. □

Throughout Sections 3–5, A is a nonempty backward-closed set. Every statement applies to each fate class realized by at least one start.

3. The two productions

Lemma 3.1 (even block). Let $m \in A$. Every even n with $m^2 \leq n < (m+1)^2$ lies in A. Writing $E(m)$ for this set, $|E(m)| \geq m$ and

$$\frac{1}{m}\Big(1 - \frac{2}{m}\Big) \ \leq\ \frac{m}{(m+1)^2} \ \leq\ \sum_{n \in E(m)} \frac{1}{n} \ \leq\ \frac{m+1}{m^2}.$$

Proof. $J(n) = \lfloor \sqrt{n} \rfloor = m$ exactly on that interval, so $n \in A$. The interval has $2m+1$ integers, at least m and at most $m+1$ of them even, each between m^2 and $(m+1)^2$. Lean: `even_block_mem`, `even_block_card`. □

Lemma 3.2 (OE fiber). Let $m \in A$ and

$$\Phi(m) = \{n \text{ odd} :\ m^4 \leq n^3 < (m+1)^4\} = \{n \text{ odd} :\ m^{4/3} \leq n < (m+1)^{4/3}\}.$$

If $n \in \Phi(m)$ and $\lfloor n^{3/2} \rfloor$ is even, then $J(J(n)) = m$, hence $n \in A$. Moreover $H_m := |\Phi(m)| \in [\frac{2}{3}m^{1/3} - 1,\ \frac{2}{3}(m+1)^{1/3} + 1]$, and the fibers of distinct m are disjoint.

Proof. $J(n) = \lfloor \sqrt{n^3} \rfloor =: k$; if k is even then $J(k) = \lfloor \sqrt{k} \rfloor$, and $\lfloor \sqrt{\lfloor \sqrt{N} \rfloor} \rfloor = m \iff m^4 \leq N < (m+1)^4$: this is the exact form of the transparent nesting $\lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor = \lfloor n^{3/4} \rfloor$. The interval $[m^{4/3}, (m+1)^{4/3})$ has length between $\frac{4}{3}m^{1/3}$ and $\frac{4}{3}(m+1)^{1/3}$ by the mean value theorem, and a half-open interval of length ℓ contains between $\ell/2 - 1$ and $\ell/2 + 1$ odd integers. Disjointness is the cell identity. Lean: `sqrt_sqrt_eq_iff`, `oe_fiber_mem`, `floorPower_oe_fiber`, `oe_fiber_disjoint`. □

Write $G_m = \#\{n \in \Phi(m) : \lfloor n^{3/2} \rfloor \text{ even}\}$. The whole argument rests on lower bounds for G_m: an elementary bound on every fiber outside a thin exceptional set (Section 4.1), and a classical average over an even block of m's (Section 4.2).

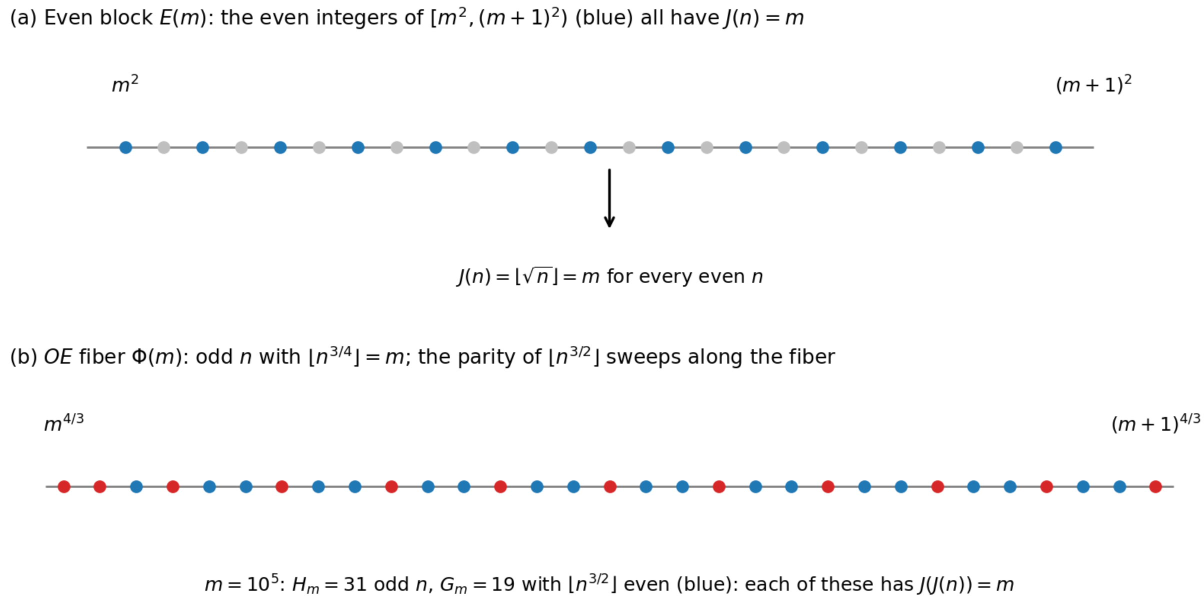

Figure 2: The two productions. (a) The even integers of $[m^2,(m+1)^2)$ all map to m. (b) The OE fiber $\Phi(m)$ for $m=10^5$: the odd n with $\lfloor n^{3/4}\rfloor=m$, coloured by the parity of $\lfloor n^{3/2}\rfloor$; those with even image (blue) have $J(J(n))=m$.

4. Parity on a fiber

4.1 The sweep lemma

On $\Phi(m)$ the quantity $x=n^{3/2}/2$ has $\lfloor n^{3/2}\rfloor$ even iff $\{x\}<\frac{1}{2}$. Consecutive odd n advance x by $\approx\frac{3}{2}n^{1/2}\approx\frac{3}{2}m^{2/3}$, and over the whole fiber this step drifts by less than one unit of $1/H_m$. So modulo 1 the fiber is an arithmetic progression with a nearly constant step α, of length H_m. Such a progression cannot avoid a half-circle unless its step is within $O(1/H_m)$ of 0, or its two-step is ($\alpha\approx\frac{1}{2}$ with the wrong phase).

Lemma 4.1 (sweep). Let $x_1<x_2<\cdots<x_H$ be real numbers with $x_{j+1}-x_j\in[a,b]$ for all j, where $0<a\le b\le\frac{1}{2}$, $b\le\frac{21}{20}a$ and $(H-1)a\ge 12$. Then

$$\#\{j:\{x_j\}<\tfrac{1}{2}\}\ \ge\ \tfrac{H}{7}\qquad\text{and}\qquad\#\{j:\{x_j\}\ge\tfrac{1}{2}\}\ \ge\ \tfrac{H}{7}.$$

The same holds with the cells $(k/2,(k+1)/2]$ in place of $[k/2,(k+1)/2)$, i.e. with the representative in $(0,1]$ in place of the fractional part.

Proof. Cut $\mathbb{R}$ into cells $C_k=[k/2,(k+1)/2)$, $k\in\mathbb{Z}$; even k are the *good* cells ($\{x\}<\frac{1}{2}$), odd k the *bad* cells. Call C_k *traversed* if $x_1\le k/2$ and $(k+1)/2\le x_H$.

(i) *A traversed cell contains at least $g:=\lfloor 1/(2b)\rfloor\ge 1$ and at most $G:=\lfloor 1/(2a)\rfloor+1$ of the points.* Let j_0 be the least index with $x_{j_0}\ge k/2$. Then $x_{j_0}<k/2+b$ (either $j_0=1$ and $x_1=k/2$, or $x_{j_0-1}<k/2$). Since every step is at most b, $x_{j_0+i}<k/2+(i+1)b\le(k+1)/2$ for all $i\le 1/(2b)-1$, and these indices exist because $(k+1)/2\le x_H$. That gives at least $\lfloor 1/(2b)\rfloor$ points in C_k. Since every step is at least a, the points in C_k are x_{j_0+i} with $k/2+ia\le x_{j_0+i}<(k+1)/2$, so $i<1/(2a)$ and there are at most $\lfloor 1/(2a)\rfloor+1$ of them. The same count holds for any cell as an upper bound.

(ii) *Counting cells.* The traversed cells are consecutive, and their number T is at least $2(x_H-x_1)-2\ge 2(H-1)a-2\ge 22$. Good and bad traversed cells alternate, so their numbers T_g,T_b differ by at most one and $T_g,T_b\ge 10$. Every point outside the traversed cells lies in the cell of x_1 or the cell of x_H.

(iii) *The ratio.* Hence Good $\geq gT_g$ and $H \leq G(T_g + T_b + 2) \leq G(2T_g + 3)$, so

$$\frac{\text{Good}}{H} \;\geq\; \frac{g}{G} \cdot \frac{T_g}{2T_g + 3} \;\geq\; \frac{g}{G} \cdot \frac{10}{23}.$$

Put $X = 1/(2a) \geq 1$, so $G = \lfloor X \rfloor + 1$ and, from $b \leq \frac{21}{20}a$, $g \geq \lfloor \frac{20}{21} X \rfloor$. If $X < 2$: $g \geq 1, G \leq 2$. If $2 \leq X < 3$: $g \geq 1, G = 3$. If $3 \leq X < 4$: $g \geq 2, G = 4$. If $4 \leq X < 5$: $g \geq 3, G = 5$. If $X \geq 5$: $g/G \geq (\frac{20}{21}X - 1)/(X + 1) \geq 0.62$. In all cases $g/G \geq \frac{1}{3}$, so Good$/H \geq \frac{10}{69} > \frac{1}{7}$. The bad count is symmetric. For left-open cells the same proof applies verbatim (the first point in a cell $(c, c + \frac{1}{2}]$ after a point $\leq c$ is $\leq c + b \leq c + \frac{1}{2}$). □

The product $g/G \cdot 10/23$ cannot reach $\frac{1}{3}$. An adversarial sequence with steps in $[a, b]$ can lock a $3 + 1$ split near $a = \frac{1}{4}$. The OE fiber is monotone.

Lemma 4.1' (monotone pairing). Keep the hypotheses of Lemma 4.1 and assume the consecutive differences are monotone (nondecreasing or nonincreasing). Then

$$\#\{j : \{x_j\} < \tfrac{1}{2}\} \;\geq\; \tfrac{H}{3} - 2 \qquad \text{and} \qquad \#\{j : \{x_j\} \geq \tfrac{1}{2}\} \;\geq\; \tfrac{H}{3} - 2.$$

The same holds with the cells $(k/2, (k+1)/2]$ in place of $[k/2, (k+1)/2)$.

Proof. The nonincreasing case is the reflection $z_j = c - x_j$, which reverses the steps and swaps the two halves. Assume the steps are nondecreasing.

Cut $\mathbb{R}$ into cells $C_k = [k/2, (k+1)/2)$ as in Lemma 4.1. An *occupied* cell contains at least one of the points. Occupied cells are consecutive; write $\rho_1, \ldots, \rho_{T^*}$ for their occupancies in order, so $\sum \rho_i = H$. Every traversed cell is occupied ($b \leq \frac{1}{2}$), hence $T^* \geq T \geq 22$. Odd-indexed cells are one colour and even-indexed cells the other.

A traversed cell of local step-scale δ has occupancy within 1 of $1/(2\delta)$. Because the steps lie in $[a, b]$ with $b \leq \frac{21}{20}a$, the values $1/(2\delta)$ vary by a factor at most $\frac{21}{20}$. Because the steps are monotone, the occupancies are a monotone sequence up to a phase error of 1: a block of L-runs, then $(L-1)$-runs, and so on (or a single block).

(i) *If $a \geq \frac{1}{4}$.* Then $X = 1/(2a) \leq 2$. Three points in a half-open cell of length $\frac{1}{2}$ would require two gaps of size at least $a \geq \frac{1}{4}$, hence a span of at least $\frac{1}{2}$, which cannot fit in $[c, c + \frac{1}{2})$. Thus every occupied cell has $\rho \in \{1, 2\}$. In the worst assignment every run of one colour has length 1 and every run of the other has length 2. If those colours have n_1 and n_2 runs, $|n_1 - n_2| \leq 1$ and $H = n_1 + 2n_2$ (or $2n_1 + n_2$). Three sub-cases: $n_1 = n_2 = n$ gives $H = 3n$ and scarcer count $n = H/3$; $n_1 = n + 1$, $n_2 = n$ gives $H = 3n + 1$ and scarcer count $n + 1 > H/3$; $n_1 = n$, $n_2 = n + 1$ gives $H = 3n + 2$ and scarcer count $n = H/3 - \frac{2}{3}$. Hence at least $\frac{H}{3} - \frac{2}{3}$.

(ii) *If $a < \frac{1}{4}$.* Pair consecutive occupied cells (ρ_{2i-1}, ρ_{2i}). Every such pair satisfies $\min(\rho, \rho') \geq (\rho + \rho')/3$. Indeed, write $L(\delta) = 1/(2\delta)$. The global drop of L is $L(a) - L(b) \leq X/21$, spread over $T^* \geq 22$ cells, so consecutive occupancies differ by at most $1 + X/441$ after the phase error. If some occupied cell has $\rho = 1$, then $g = 1$, hence $\frac{20}{21}X < 2$ and $X < 2.1$; the drop of L is then < 0.1, consecutive occupancies differ by at most 1, and the pairs are $(1, 1)$, $(1, 2)$ or $(2, 2)$, each with min $\geq$ (sum)$/3$. If every occupancy is at least 2, a consecutive pair is (k, k), $(k, k+1)$, or $(k, k+d)$ with $k \geq 2$ and $d \leq 1 + X/441$; then $\min/\text{sum}$ is $\frac{1}{2}$, at least $\frac{2}{5}$, or $k/(2k + d) \geq \frac{1}{3}$ (the last because $k \geq 2$ and, when $d = 2$, $k/(2k+2) = \frac{1}{3}$; when X is large enough for $d \geq 3$, one has $k \geq g \geq \frac{20}{21}X - 1$ and $k/(2k + d) \geq \frac{1}{2} - o(1) > \frac{1}{3}$).

Thus each pair contributes at least one-third of its points to each colour. If T^* is odd the leftover cell has at most G points, so the scarcer colour has at least $(H - G)/3$ points.

- If $G \leq 6$, this is at least $\frac{H}{3} - 2$.
- If $G \geq 7$, then $X \geq 6$ and $g \geq \lfloor \frac{20}{21} \cdot 6 \rfloor = 5$. Every pair then has $\min/\text{sum} \geq \frac{2}{5}$: either $X < 441$, so consecutive difference at most 2 and the worst pair is $(5, 7)$ with ratio $\frac{5}{12}$, or $X \geq 441$ and $g/(2g + 1 + X/441) \geq \frac{2}{5}$. Hence the scarcer count is at least $\frac{2}{5}(H - G)$. The hypothesis $(H - 1)a \geq 12$

gives $H \geq 24X+1 \geq 24(G-1)+1$, and $\frac{2}{5}(H-G) \geq \frac{H}{3}-2$ follows: $\frac{1}{15}H \geq \frac{2}{5}G-2$, i.e. $H \geq 6G-30$, which holds because $24G-23 \geq 6G-30$.

In all cases the scarcer count is at least $\frac{H}{3}-2$. The two colours are the two half-interval counts. For left-open cells the same argument applies with the cells $(k/2,(k+1)/2]$. □

Lemma 4.2 (fiber parity). For $m \geq 10^6$ put $\alpha_m = \{\frac{3}{2}m^{2/3}\}$ and call m *good* if

$$\|\alpha_m\| \;\geq\; 22\,m^{-1/3} \qquad \text{and} \qquad \|\alpha_m - \tfrac{1}{2}\| \;\geq\; 2\,m^{-1/3},$$

where $\|\cdot\|$ is the distance to the nearest integer. If m is good then $G_m \geq \frac{1}{3}H_m-2$ and $H_m-G_m \geq \frac{1}{3}H_m-2$.

Proof. Let $n_1 < \cdots < n_H$ be the odd integers of $\Phi(m)$, $n_{j+1}=n_j+2$, and $x_j = n_j^{3/2}/2$. Then $\lfloor n_j^{3/2}\rfloor$ is even iff $\{x_j\} < \frac{1}{2}$. The steps are

$$\delta_j = x_{j+1}-x_j = \int_{n_j}^{n_j+2} \tfrac{3}{4}t^{1/2}\,dt \in \big[\tfrac{3}{2}n_j^{1/2},\ \tfrac{3}{2}(n_j+2)^{1/2}\big] \subseteq [A_m, B_m],$$

with $A_m = \frac{3}{2}m^{2/3}$ and $B_m = \frac{3}{2}\big((m+1)^{4/3}+2\big)^{1/2}$. Using $\sqrt{u}-\sqrt{v} \leq (u-v)/(2\sqrt{v})$ and $(m+1)^{4/3}-m^{4/3} \leq \frac{4}{3}(m+1)^{1/3}$,

$$\eta_m := B_m - A_m \leq \frac{(m+1)^{1/3}+\frac{3}{2}}{m^{2/3}} \leq m^{-1/3}\Big(1+\frac{1}{3m}+\frac{3}{2}m^{-1/3}\Big) \leq 1.02\,m^{-1/3}.$$

Also $H_m - 1 \geq \frac{2}{3}m^{1/3}-2 \geq 0.646\,m^{1/3}$.

Case 1: $\alpha_m \leq \frac{1}{2}-2m^{-1/3}$. Put $N=\lfloor A_m\rfloor$ and $y_j = x_j - jN$; then $\{y_j\}=\{x_j\}$ and the steps of y lie in $[a,b]=[\alpha_m,\alpha_m+\eta_m] \subseteq [22m^{-1/3},\frac{1}{2}]$. Now $b/a \leq 1+1.02/22 < \frac{21}{20}$ and $(H-1)a \geq 0.646\cdot 22 > 12$. The steps of y are nondecreasing. Lemma 4.1' applies.

Case 2: $\alpha_m \geq \frac{1}{2}+2m^{-1/3}$. Put $z_j = j(N+1)-x_j$, increasing with steps $(N+1)-\delta_j \in [1-\alpha_m-\eta_m,\ 1-\alpha_m] \subseteq [20.98\,m^{-1/3},\ \frac{1}{2}]$; again $b/a \leq 1+1.02/20.98 < \frac{21}{20}$ and $(H-1)a \geq 0.646\cdot 20.98 > 12$. Write $\langle z\rangle \in (0,1]$ for the representative of z modulo 1 in $(0,1]$. Then $\langle z_j\rangle = 1-\{x_j\}$ for every j, so $\{x_j\} < \frac{1}{2} \iff \langle z_j\rangle \in (\frac{1}{2},1]$. The steps of z are monotone. Lemma 4.1' in its left-open form gives both counts.

The middle range $|\alpha_m - \frac{1}{2}| < 2m^{-1/3}$ and the range $\|\alpha_m\| < 22m^{-1/3}$ are excluded by goodness. □

Lemma 4.3 (bad fibers are thin). For $u \geq 10^6$,

$$\#\{m \in (u,2u] :\ m \text{ bad}\} \leq 63\,u^{2/3}, \qquad \sum_{\substack{m>U\\ m \text{ bad}}} \frac{1}{m} \leq 306\,U^{-1/3} \quad (U \geq 10^6).$$

Proof. $\varphi(m) = \frac{3}{2}m^{2/3}$ increases with $\varphi(m+1)-\varphi(m) \in [(m+1)^{-1/3}, m^{-1/3}] \subseteq [(3u)^{-1/3}, u^{-1/3}]$ on $(u,2u]$. Badness means $\{\varphi(m)\}$ lies in one of two arcs of the circle, of total length at most $48\,u^{-1/3}$. As m runs over $(u,2u]$, φ increases by $\frac{3}{2}u^{2/3}(2^{2/3}-1) < 0.882\,u^{2/3}$, so it passes each arc at most $0.882u^{2/3}+2$ times, and each pass through an arc of length w takes at most $w(3u)^{1/3}+1$ values of m. Hence the count is at most $(0.882u^{2/3}+2)(48\cdot 3^{1/3}+2) \leq 63\,u^{2/3}$ for $u \geq 10^6$. Summing $63\,(2^iU)^{-1/3}$ over $i \geq 0$ gives $63U^{-1/3}/(1-2^{-1/3}) \leq 306\,U^{-1/3}$. □

4.2 The block average

The adversarial sweep $G_m \geq H_m/7$ is not the fiber bound: Lemma 4.1' gives the monotone floor $H_m/3-2$. In the numerical experiments of Section 11 the mean of G_m/H_m is $\frac{1}{2}$ and the minimum on good fibers is near $\frac{1}{3}$. When the members of A come in even blocks — as the E-produced part of A always does — the fibers can be averaged over the block, and the average is exactly $\frac{1}{2}$ by a classical exponential-sum estimate.

Proposition 4.4 (block average). For $m' \geq 2$ let $I(m') = [m'^{8/3}, (m'+1)^{8/3})$ and

$$U(m') = \{n \text{ odd in } I(m') : \lfloor n^{3/4} \rfloor \text{ even}, \lfloor n^{3/2} \rfloor \text{ even}\} = \bigsqcup_{m \in E(m')} \{n \in \Phi(m) : \lfloor n^{3/2} \rfloor \text{ even}\}.$$

With the explicit constant $C_0 = 250$,

$$|U(m')| \ \geq \ \tfrac{1}{4} \#\{n \text{ odd in } I(m')\} - 250\, m'^{11/9} \log(m'+1).$$

Since $\#\{n \text{ odd in } I(m')\} \geq \frac{4}{3} m'^{5/3} - 1$, this reads $|U(m')| \geq \frac{1}{3} m'^{5/3}(1 - \varepsilon_B(m'))$ with $\varepsilon_B(m') = \frac{3}{4} m'^{-5/3} + 750\, m'^{-4/9} \log(m'+1) \to 0$.

Proof. Write $\psi(y) = (-1)^{\lfloor y \rfloor}$, so that $[\lfloor y \rfloor \text{ even}] = \frac{1}{2}(1 + \psi(y))$, and

$$|U(m')| = \tfrac{1}{4} \sum_n \big(1 + \psi(n^{3/4}) + \psi(n^{3/2}) + \psi(n^{3/4})\psi(n^{3/2})\big),$$

the sum over odd $n \in I(m')$. Four terms.

Term 1 is the main term.

Term 2. $\lfloor n^{3/4} \rfloor = m$ is constant on each fiber, so $\sum \psi(n^{3/4}) = \sum_{m'^2 \leq m < (m'+1)^2} (-1)^m H_m$. Since H_m is within 1 of half the fiber length and consecutive fiber lengths differ by at most 1, $|H_{m+1} - H_m| \leq 3$; pairing consecutive m gives $|\text{Term 2}| \leq 3m' + \max H_m \ll m'$.

Term 3. Expand ψ by Vaaler's theorem [10], in the form of Paper B, Lemma 3.5: $|\psi(y) - V_J(y/2)| \leq \Delta_J(y/2)$, where $V_J(t) = \sum_{0<|q|\leq J} a_q e(qt)$ with $|a_q| \leq \min(1, 2/|q|)$ and $\Delta_J \geq 0$ is a trigonometric polynomial of degree J with constant term and coefficients at most $1/(J+1)$. Reindex $n = 2r+1$. For $q \neq 0$ the phase $f(r) = q(2r+1)^{3/2}/2$ has $f''(r) = \frac{3}{2} q\, n^{-1/2}$, which on $I(m')$ lies between $\lambda_q = \frac{3}{2}|q|(m'+1)^{-4/3}$ and $\alpha\lambda_q$ with $\alpha = 2^{4/3}$ (since $(1+1/m')^{4/3} \leq 2^{4/3}$ for $m' \geq 1$). An explicit form of the second-derivative test (van der Corput; [13, Theorem 2.2]) with interval length $M \leq \frac{4}{3}(m'+1)^{5/3} + 1$ is

$$\Big|\sum_r e(f(r))\Big| \ \leq \ 8\sqrt{\alpha}(M\lambda_q^{1/2} + \lambda_q^{-1/2}) \ = \ A_3 |q|^{1/2} + B_3 |q|^{-1/2},$$

where $A_3 = 8\sqrt{\alpha}\, M \sqrt{\frac{3}{2}}\, (m'+1)^{-2/3}$ and $B_3 = 8\sqrt{\alpha}\sqrt{\frac{2}{3}}\, (m'+1)^{2/3}$. Summing against $|a_q| \leq 2/|q|$ over $0 < |q| \leq J$ gives at most $4\sum_{q \leq J}(A_3 q^{-1/2} + B_3 q^{-3/2}) \leq 4A_3(1 + 2J^{1/2}) + 4B_3\zeta(\frac{3}{2})$, with $\zeta(\frac{3}{2}) < 2.613$. The Δ_J error is at most $M/(J+1) + (2/(J+1))\sum_{q \leq J}(A_3 q^{1/2} + B_3 q^{-1/2})$. With $J = \max(1, \lfloor m'^{4/9} \rfloor)$ both contributions are $O(m'^{11/9})$, with an explicit majorant recorded at the end of the proof.

Term 4. Expand both factors. The main part is $\sum_{q_1, q_2 \neq 0} a_{q_1} a_{q_2} \sum_r e(f_{q_1,q_2}(r))$ with $f_{q_1,q_2}(r) = \frac{1}{2}(q_1(2r+1)^{3/4} + q_2(2r+1)^{3/2})$ and

$$f''_{q_1,q_2}(r) = \tfrac{3}{2} q_2 n^{-1/2}\Big(1 - \frac{q_1}{4q_2}\, n^{-3/4}\Big).$$

For $|q_1| \leq J = \max(1, \lfloor m'^{4/9} \rfloor)$ and $n \geq m'^{8/3}$ the correction is $\leq \frac{1}{4} m'^{4/9-2} \leq \frac{1}{16}$ for $m' \geq 2$, so $|f''|$ lies between λ_{q_2} and $\alpha\lambda_{q_2}$ up to a factor $\frac{17}{16}$, uniformly in q_1; we absorb this by replacing $8\sqrt{\alpha}$ with $8\sqrt{\alpha} \cdot \frac{17}{16}$ in Term 4 only. The second-derivative test then gives the same shape as in Term 3, and the double sum against $|a_{q_1} a_{q_2}| \leq 4/(|q_1||q_2|)$ is at most $16(\log J + 1)\sum_{q \leq J}(A_3 q^{-1/2} + B_3 q^{-3/2})$. Three error terms remain. $\sum \Delta_J(n^{3/2}/2)$ is bounded as in Term 3. For $\sum \Delta_J(n^{3/4}/2)$ the constant term gives $M/(J+1)$ and each mode $0 < |q_1| \leq J$ gives $(J+1)^{-1}|\sum_r e(q_1(2r+1)^{3/4}/2)|$; here the phase has derivative $\frac{3}{4} q_1 n^{-1/4}$, monotone and of size between $\frac{3}{4}|q_1|(m'+1)^{-2/3}$ and $\frac{3}{4} m'^{4/9-2/3} < \frac{1}{2}$, so the Kusmin–Landau first-derivative test [13, Theorem 2.1] bounds the sum by $1/\lambda \leq \frac{4}{3}(m'+1)^{2/3}/|q_1|$ (using $\cot(\pi\lambda) \leq 1/\lambda$); the modes contribute at most $\frac{8}{3}(m'+1)^{2/3}(\log J + 1)/(J+1)$ in total. The product error $\sum \Delta_J \Delta_J$ is at most $\sup \Delta_J \leq 2$ times either of the two previous sums.

Term 2, explicitly. Pairing consecutive fibres gives $|\text{Term 2}| \le 3m' + \max H_m$, and $\max H_m \le \frac{2}{3}(m'+1)^{2/3}+1$, so $|\text{Term 2}| \le 3m' + \frac{2}{3}(m'+1)^{2/3}+1$.

Collecting, $|U(m') - \frac{1}{4}\#\{n \text{ odd in } I(m')\}|$ is at most one-quarter of the sum of the four term-bounds. Substituting the displays, the resulting majorant divided by $m'^{11/9}\log(m'+1)$ equals 217 at $m'=2$ and is smaller for every larger m' (the large-m' limit is < 90, dominated by the Term-4 harmonic sum). We take $C_0 = 250$. $\square$

The proposition is not sharp — the census of Section 11 sits at 0.078 of the same scale — but $C_0 = 250$ is an explicit proved constant, and it is all the recursion needs.

5. The recursion and the contagion theorem

5.1 Three sources of members of A in $(\sqrt{x}, x]$

Let $x \ge 2$ and $t = \log x$. Three families of members of $A \cap (\sqrt{x}, x]$ are pairwise disjoint.

1. **E-images.** For $m \in A \cap (x^{1/4}, \sqrt{x}-1]$, $E(m) \subseteq (\sqrt{x}, x]$ (as $m^2 > \sqrt{x}$ and $(m+1)^2 \le x$). Distinct m give disjoint blocks. By Lemma 3.1 the log-mass is at least $\sum_m \frac{1}{m}(1-\frac{2}{m}) \ge (1-2x^{-1/4})\,\ell_A(x^{1/4}, \sqrt{x}-1]$, and $\ell_A(x^{1/4}, \sqrt{x}-1] \ge g_A(t/2) - 2x^{-1/2}$ (at most one integer lies in $(\sqrt{x}-1, \sqrt{x}]$).
2. **OE-images of E-blocks.** For $m' \in A \cap (x^{3/16}, x^{3/8}-1]$, $U(m') \subseteq A \cap (\sqrt{x}, x]$ (Lemma 3.2 applied to the even $m \in E(m') \subseteq A$; $I(m') \subseteq (\sqrt{x}, x]$ as $m'^{8/3} > \sqrt{x}$ and $(m'+1)^{8/3} \le x$). Distinct m' give disjoint $I(m')$. Each $n \in U(m')$ has $1/n > (m'+1)^{-8/3}$, so by Proposition 4.4 the log-mass of $U(m')$ is at least $\frac{1}{3}m'^{5/3}(1-\varepsilon_B(m'))(m'+1)^{-8/3} \ge \frac{1}{3m'}(1-\varepsilon_B(m') - \frac{8}{3m'})$, and summing, $\ge \frac{1}{3}(1-\varepsilon'(x))(g_A(3t/8) - 2x^{-3/8})$ with $\varepsilon'(x) = \sup_{m' > x^{3/16}}(\varepsilon_B(m') + \frac{8}{3m'}) \to 0$.
3. **OE-images of the rest.** Let $A_x^E = \bigcup_{m'' \in A} E(m'') \cap (x^{3/8}, x^{3/4}]$ be the E-produced part of $A \cap (x^{3/8}, x^{3/4}]$ and A_x^{rest} its complement in $A \cap (x^{3/8}, x^{3/4}]$. The blocks meeting $(x^{3/8}, x^{3/4}]$ come from $m'' \in [x^{3/16}-1, x^{3/8}]$, so $\ell(A_x^E) \le \sum_{m''} \frac{m''+1}{m''^2} \le (1+2x^{-3/16})(g_A(3t/8) + 2x^{-3/16})$ and $\ell(A_x^{\text{rest}}) \ge g_A(3t/4) - (1+2x^{-3/16})(g_A(3t/8)+2x^{-3/16})$. For $m \in A_x^{\text{rest}}$ with $m \le x^{3/4}-1$ (this drops at most one element, of log-mass $\le 2x^{-3/4}$), $\Phi(m) \subseteq (\sqrt{x}, x]$; the fibers are disjoint from each other and from those of item 2 (different m), and for good $m \ge 10^6$ Lemmas 4.2 and 3.2 give log-mass at least $(\frac{1}{3}H_m - 2)(m+1)^{-4/3} \ge \frac{2}{9m}(1 - O(m^{-1/3}))$. Bad $m > x^{3/8}$ carry log-mass at most $306x^{-1/8}$ (Lemma 4.3). Hence item 3 contributes at least $\frac{2}{9}(1-2x^{-1/8})[\ell(A_x^{\text{rest}}) - 306x^{-1/8} - 2x^{-3/4}]_+$.

The images in item 1 are even, those in items 2–3 odd, so the three families are disjoint, and all lie in $(\sqrt{x}, x]$.

5.2 The functional inequalities

Adding items 1 and 2:

$$g_A(t) \;\ge\; (1-\varepsilon_1(t))\, g_A(t/2) + \tfrac{1}{3}(1-\varepsilon_2(t))\, g_A(3t/8) - \varepsilon_3(t), \tag{5.1}$$

with $\varepsilon_1 = 2e^{-t/4}$, $\varepsilon_2 = \varepsilon'(e^t)$, $\varepsilon_3 = 2e^{-t/2} + e^{-3t/8}$, all tending to 0. Adding item 3 as well, and noting that the coefficient of the *actual* value $g_A(3t/8)$ is $\frac{1}{3}(1-\varepsilon_2) - \frac{2}{9}(1+2e^{-3t/16}) \to \frac{1}{9} > 0$:

$$g_A(t) \;\ge\; (1-\varepsilon_1)\, g_A(t/2) + (\tfrac{1}{9} - \varepsilon_4) g_A(3t/8) + (\tfrac{2}{9} - \varepsilon_5) g_A(3t/4) - \varepsilon_6(t), \tag{5.2}$$

with $\varepsilon_4, \varepsilon_5, \varepsilon_6 \to 0$ (for t so large that $x^{3/8} \ge 10^6$; the bracket $[\cdot]_+$ may be dropped because the right side of (5.2) is a lower bound for it). Every coefficient of a g_A-value on the right is nonnegative for large t, so lower bounds for g_A at $t/2$, $3t/8$, $3t/4$ transfer.

5.3 The recursion lemma

The passage from a functional inequality with vanishing errors to a power lower bound is a standalone statement.

Lemma 5.1 (recursion lemma). Let $e_1, \dots, e_r \in (0,1)$, $c_1, \dots, c_r \ge 0$, and $\lambda \in (0,1)$ with

$$\zeta := \sum_{i=1}^{r} c_i e_i^{\lambda} - 1 > 0.$$

Put $e_{\min} = \min_i e_i$, $e_{\max} = \max_i e_i$. Let $g : (0,\infty) \to [0,\infty)$ satisfy, for all $t \ge t_1$,

$$g(t) \;\ge\; \sum_{i=1}^{r} (c_i - \eta_i(t))\, g(e_i t) - \eta_0(t),$$

where $0 \le \eta_i(t) \le c_i$ for $i \ge 1$, $\eta_0(t) \ge 0$, and $\sum_{i\ge 1} \eta_i(t) e_i^{\lambda} \le \zeta/3$ and $\eta_0(t) \le \frac{2\zeta}{3} c_0$ for all $t \ge t_1$, for some $c_0 > 0$ with $g(t) \ge c_0$ on $[e_{\min} t_1,\, t_1]$. Then

$$g(t) \;\ge\; K\, t^{\lambda} \quad \text{for all } t \ge e_{\min} t_1, \qquad K = c_0 t_1^{-\lambda}.$$

Proof. On $[e_{\min} t_1, t_1]$, $Kt^{\lambda} \le K t_1^{\lambda} = c_0 \le g(t)$. Suppose $g(t) \ge Kt^{\lambda}$ holds on $[e_{\min} t_1, T]$ for some $T \ge t_1$, and let $t \in (T, T/e_{\max}]$. Then $e_i t \le e_{\max} t \le T$ and $e_i t \ge e_{\min} t > e_{\min} t_1$, so each $e_i t$ lies in the interval where the bound is known, and $t \ge t_1$, so

$$g(t) \;\ge\; \sum_i (c_i - \eta_i(t)) K (e_i t)^{\lambda} - \eta_0(t) \;=\; Kt^{\lambda}\Big[\sum_i c_i e_i^{\lambda} - \sum_i \eta_i(t) e_i^{\lambda}\Big] - \eta_0(t) \;\ge\; Kt^{\lambda}\Big(1 + \tfrac{2\zeta}{3}\Big) - \tfrac{2\zeta}{3} c_0 \;\ge\; Kt^{\lambda},$$

using $c_0 = K t_1^{\lambda} \le K t^{\lambda}$. Hence the bound holds on $[e_{\min} t_1, T/e_{\max}]$; induction on N covers $[e_{\min} t_1, t_1 e_{\max}^{-N}]$ for every N, i.e. all $t \ge e_{\min} t_1$. □

5.4 The seed

Lemma 5.2 (seed). Every nonempty backward-closed A contains an integer $m \ge 3$, and for any such m,

$$g_A(t) \;\ge\; c_A := \Big(1 - \frac{2}{m^4}\Big)\Big(\frac{0.375}{m+1} - \frac{1}{(m+1)^2 - 1}\Big) > 0 \qquad \text{for all } t \ge 4\log(m+1).$$

Proof. If $A \ni a \le 2$ then $E(a) \subseteq A$ contains 2 or 4, and $E(2) \ni 4$, so $A \ni 4$. Fix $m \ge 3$ in A and let $S_1 = E(m)$, $S_{k+1} = \bigcup_{m'' \in S_k} E(m'')$. Then $S_k \subseteq A \cap [m^{2^k}, (m+1)^{2^k})$ and, by Lemma 3.1, $\ell(S_{k+1}) \ge (1 - 2m^{-2^k})\ell(S_k)$, so $\ell(S_k) \ge \ell(S_1) \prod_{j\ge 1}(1 - 2m^{-2^j}) \ge 0.75\, \ell(S_1) \ge \frac{0.375}{m+1}$ for $m \ge 3$. Let $y \ge (m+1)^4$ and choose $k \ge 2$ with $(m+1)^{2^k} \le y < (m+1)^{2^{k+1}}$. Then $S_k \subseteq (y^{1/4}, y]$ (as $y < m^{2^{k+2}}$). The members of S_k above $\sqrt{y}$ lie in $(\sqrt{y}, y]$; the members in $(y^{1/4}, \sqrt{y} - 1]$ have their E-images in $(\sqrt{y}, y]$, with log-mass at least $(1 - 2m^{-2^k})$ times their own; at most one member lies in $(\sqrt{y} - 1, \sqrt{y}]$, with $1/n \le 1/((m+1)^2 - 1)$. Hence $g_A(\log y) \ge (1 - 2m^{-4})(\ell(S_k) - 1/((m+1)^2 - 1)) \ge c_A$, and $c_A > 0$ because $0.375(m+1) \ge 1 + 0.375/(m+1)$ for $m \ge 3$. □

5.5 The theorem

Let λ^* and λ^{**} be the roots in $(0,1)$ of

$$2^{-\lambda} + \tfrac{1}{3}\big(\tfrac{3}{8}\big)^{\lambda} = 1, \qquad 2^{-\lambda} + \tfrac{1}{9}\big(\tfrac{3}{8}\big)^{\lambda} + \tfrac{2}{9}\big(\tfrac{3}{4}\big)^{\lambda} = 1.$$

Numerically $\lambda^* = 0.3774\ldots$ and $\lambda^{**} = 0.4480\ldots$. For comparison: the elementary sweep alone, without Proposition 4.4, gives $2^{-\lambda} + \frac{2}{21}(\frac{3}{4})^{\lambda} = 1$, root $0.138\ldots$; perfect fiber equidistribution at this depth would give $2^{-\lambda} + \frac{1}{3}(\frac{3}{4})^{\lambda} = 1$, root $\lambda_{\text{ideal}} = 0.4927\ldots$, the ceiling of the depth-two method.

Theorem 5.3 (logarithmic density of a backward-closed set). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $0 < \lambda < \lambda^{**}$. There are $c > 0$ and x_0, depending on A and λ, such that

$$\sum_{\substack{n \in A \\ n \le x}} \frac{1}{n} \;\ge\; c\,(\log x)^{\lambda} \qquad \text{for all } x \ge x_0.$$

The same conclusion for $\lambda < \lambda^*$ uses only Proposition 4.4 (no sweep lemma).

Proof. Apply Lemma 5.1 to $g = g_A$ with $(e_i, c_i) = (\frac{1}{2}, 1), (\frac{3}{8}, \frac{1}{9}), (\frac{3}{4}, \frac{2}{9})$, so that $\zeta = 2^{-\lambda} + \frac{1}{9}(\frac{3}{8})^\lambda + \frac{2}{9}(\frac{3}{4})^\lambda - 1 > 0$ exactly when $\lambda < \lambda^{**}$. Inequality (5.2) has the required form with $\eta_1 = \varepsilon_1$, $\eta_2 = \varepsilon_4$, $\eta_3 = \varepsilon_5$, $\eta_0 = \varepsilon_6$, all tending to 0. Let $m \geq 3$ be a member of A and $c_0 = c_A$ from Lemma 5.2. Choose t_1 so large that $\frac{3}{8}t_1 \geq 4\log(m+1)$, that $e^{3t_1/8} \geq 10^6$, and that for all $t \geq t_1$ the errors satisfy $\sum_i \eta_i(t) e_i^\lambda \leq \zeta/3$ and $\eta_0(t) \leq \frac{2\zeta}{3} c_A$. Lemma 5.2 gives $g_A \geq c_A$ on $[\frac{3}{8}t_1, t_1]$. Lemma 5.1 then gives $g_A(t) \geq K t^\lambda$ for all $t \geq \frac{3}{8}t_1$, and $\sum_{n \in A,\, n \leq x} 1/n \geq g_A(\log x)$. For $\lambda < \lambda^*$ use (5.1) with $(e_i, c_i) = (\frac{1}{2}, 1), (\frac{3}{8}, \frac{1}{3})$. □

Corollary 5.4 (natural density, infinitely often). For every $\lambda < \lambda^{**}$ there is $c > 0$ such that for every $X \geq x_0$ some $y \in (\sqrt{X}, X]$ satisfies $\#(A \cap (y/2, y]) \geq c\, y\, (\log y)^{\lambda - 1}$.

Proof. $\ell_A(\sqrt{X}, X] = g_A(\log X) \geq K(\log X)^\lambda$ is spread over at most $\frac{1}{2}\log_2 X + 2$ dyadic blocks $(y/2, y] \subseteq (\sqrt{X}, 2X]$; one of them has $\ell_A(y/2, y] \geq K'(\log X)^{\lambda-1}$, hence $\#(A \cap (y/2, y]) \geq \frac{y}{2}\, \ell_A(y/2, y] \geq \frac{K'}{2}\, y\, (2\log y)^{\lambda-1}$, because $\log X \leq 2 \log y$ and $\lambda - 1 < 0$. □

Corollary 5.5 (fate contagion). Let φ be a fate realized by at least one start: the trivial cycle, a particular nontrivial cycle C, or divergence. Then $\{n : \mathrm{fate}(n) = \varphi\}$ satisfies Theorem 5.3 and Corollary 5.4. In particular:

1. $\sum_{n \in R,\ n \leq x} 1/n \gg (\log x)^\lambda$ for every $\lambda < \lambda^{**}$.
2. If some start does not reach 1, then $\sum_{n \notin R,\ n \leq x} 1/n \gg (\log x)^\lambda$, and on infinitely many dyadic blocks the failures have natural density $\gg (\log y)^{\lambda - 1}$.
3. If a nontrivial cycle exists, the set of starts that enter it has the same lower bounds; if a divergent orbit exists, so does the set of divergent starts.

Proof. Lemma 2.1 and Theorem 5.3. □

5.6 Remarks

Why the exponent is not 1*.* Not because the method is capped below 1 — Section 5.7 shows its ceiling is exactly 1 — but because two named obstructions stop the production list short of it. Positive density of R is therefore as hard as the full almost-all-descent statement with a rate, and is not claimed.

Pointwise natural density. Theorem 5.3 is logarithmic and Corollary 5.4 holds infinitely often, not for every x. A pointwise bound $\#(A \cap [1, x]) \gg x(\log x)^{\lambda-1}$ cannot be inducted through fixed-ratio intervals — the E-preimage of a ratio-r interval has ratio $\sqrt{r}$ — and for a backward-closed set generated by a single seed it is false: the E-tree of one integer is lacunary at the dyadic scale.

Sharpness of the constants. The monotone pairing gives $\frac{1}{3}H_m - 2$; the observed floor on good fibers is 0.328 at $\alpha_m \approx \frac{1}{3}$ (Section 11). The remaining depth-two gap $0.448 \to 0.4927$ is a dynamical averaging problem for the low-even set $P = \{m : G_m/H_m \leq 0.40\}$, not another pointwise fiber bound. By Proposition 5.13 it is not a separate problem either: it is exactly the short-fiber regime $\rho_w > \frac{1}{2}$. Section 5.7 identifies it further, and more sharply: the gap is closed **exactly** by an infinite family of elementary productions, so it need not be read as a dynamical averaging problem at all.

What is excluded. Nothing. Corollary 5.5 does not say that a cycle or a divergent orbit is impossible; it says that either would be common.

5.7 The ceiling of the production calculus

Every production used above is a parity word w applied backwards. Write $a = \#\{E \text{ in } w\}$, $b = \#\{O \text{ in } w\}$. The source of a w-production sits at log-scale $\rho_w t$ with

$$\rho_w = 2^{-a}\left(\tfrac{3}{2}\right)^b, \qquad c_w^{\text{ideal}} = \frac{2^{-|w|}}{\rho_w}, \tag{5.7}$$

the second being the log-mass coefficient when the fiber realizes the fair share $2^{-|w|}$. Formula (5.7) returns every coefficient in this section and in Appendix C: $c_E = 1$, $c_{OE} = c_{OEE} = \frac{1}{3}$, $c_{OOEEE} = \frac{1}{9}$.

Proposition 5.12 (the ceiling is 1). Let the E-production be exact and let every O-production realize a fraction η of the ideal fiber share. Then the recursion exponent is the root of

$$2^{-\lambda} + \tfrac{\eta}{3}\big(\tfrac{3}{2}\big)^{\lambda} = 1, \tag{5.8}$$

so $\lambda = 1$ exactly when $\eta = 1$, and $d\lambda/d\eta|_{\eta=1} = 1/\ln(4/3) = 3.4761\ldots$.

Proof. Summing the backward tree by path type — odd share θ, $\binom{k}{\theta k}$ arrangements, mass $(\eta/3)^{\theta k}$, scale constraint $k(1-\theta\log_2 3) \le u$ — gives $\lambda(\eta) = \max_\theta[H(\theta) - \theta\log_2(3/\eta)]/[1-\theta\log_2 3]$, whose stationary point is $\theta = \frac{1}{2}$; there numerator and denominator are both $1 - \frac{1}{2}\log_2 3$ when $\eta = 1$, so $\lambda = 1$. Equivalently $\frac{1}{2} + \frac{1}{3}\cdot\frac{3}{2} = 1$ in (5.8). Differentiating (5.8) at $(\lambda, \eta) = (1, 1)$ gives $-\partial_\lambda = \frac{1}{2}\ln\frac{4}{3}$ and $\partial_\eta = \frac{1}{2}$. □

The extremal backward path is therefore the *fair-coin* one, $\theta = \frac{1}{2}$: the ceiling of the method is exactly the value the frontier statement of Section 12 needs, not less. Two obstructions keep the realized η below 1.

Proposition 5.13 (fiber-length criterion). For a production word w, the fiber over a source m' is $I(m') = [m'^{1/\rho_w}, (m'+1)^{1/\rho_w})$, so at scale $P = n$

$$|I(m')| = \tfrac{1}{\rho_w}\, m'^{1/\rho_w - 1} = \tfrac{1}{\rho_w}\, P^{1-\rho_w}.$$

The localized parity estimate of Appendix C requires $|I| \ge P^{1/2}$. Hence the ideal share is available **exactly when** $\rho_w \le \frac{1}{2}$.

This classifies every production: E ($\rho = \frac{1}{2}$, exact), *OEE* ($\rho = \frac{3}{8}$, $|I| = P^{5/8}$, block average, ideal), *OOEEE* ($\rho = \frac{9}{32}$, $|I| = P^{23/32}$, Appendix C, ideal) — against *OE* ($\rho = \frac{3}{4}$, $|I| = P^{1/4}$), the one lossy production, and *OOEE* ($\rho = \frac{9}{16}$, $|I| = P^{7/16}$), which is why Appendix C must reach *OOEEE*: the extra E lengthens the fiber past the threshold.

The K_3 cap. Production words must be prefix-free, so the natural list is the first passage of ρ_w below $\frac{1}{2}$. A run of r consecutive O's forces r nested 3/2-powers plus the closing square root — Paper B at depth $r+1$. Paper B is complete to depth 4, and to depth 5 except $OOOO*$, which is the parked kernel K_3. Hence $r \le 3$, and with $\lambda(r)$ the exponent when O-runs up to length r are controlled:

r	Paper B depth	$\lambda(r)$, present sweep	$\lambda(r)$, ideal fibers
1	2	0.4480	0.4927
2	3	0.6247	0.7180
3	4	0.7095	0.8414
4	5	0.7516	0.9121

Row $r = 4$ is behind K_3. Appendix C uses one word of the $r = 2$ family and prints 0.5392; its Proposition C.2 attains the *ideal* share, so the ground between 0.5392 and 0.7180 is more words — each costing one localization of a Paper B theorem to sub-dyadic intervals — and not a better estimate for the word already used. $\lambda = 1$ needs $r \to \infty$: **the contagion side of the problem is guarded by K_3 as well**, at $r = 4$, and approaches the same wall from the other direction rather than going around it.

The $r = 1$ words cost nothing. The table over-charges every word with no two consecutive O's. In such a word each isolated O is absorbed by the E after it: the transparent nesting of Lemma 3.2 makes $J^2(n) = \lfloor n^{3/4}\rfloor$ an *exact* function of n, so every later parity is a function of that one integer w, on whose level sets — intervals $[w^{4/3}, (w+1)^{4/3})$ of length $\asymp P^{1/4}$ — those parities are constant, and in which the surviving ψ's are ψ of smooth monomials rather than of nested floors. Only $\psi(n^{3/2})$ varies inside a block. For *OEOEE* ($\rho = \frac{9}{32}$, the same fiber length as Appendix C's *OOEEE*) this splits the fifteen sign sums into eight that are one-variable sums in w, handled by the second-derivative test and the Proposition 4.4 pairing, and seven that are $\sum_w |T(w)| \le \mathcal{N}^{1/2}(\sum_w |T(w)|^2)^{1/2}$ with $T(w) = \sum_{n\in I_w} \psi(n^{3/2})$, handled by Kusmin–Landau per block against a monotone slowly-stepping frequency. The savings are $P^{-3/32}$ and $P^{-1/16}$; no Paper B estimate, no localization, and — unlike $OOEEE$, where $\lfloor v^{1/4}\rfloor = \lfloor n^{9/16}\rfloor$ fails on a set needing Erdős–Turán — no exceptional set: the fiber $J^5(n) = m' \iff \lfloor n^{3/4}\rfloor \in [m'^{8/3}, (m'+1)^{8/3})$ is exact. Since an $OEOEE$-start lies in the OE-fiber of an *odd* $J^2(n)$, the family sits inside family 3 and must be removed from it before being re-added at the ideal share, which by the two-sided form of Proposition 4.4 changes (5.2) by exactly $+\frac{1}{27}g_A(9t/32)$: λ^{**} would rise to 0.4801 unconditionally and λ^{***} to 0.5665.

The construction iterates. Write a no-OO word as $(OE)^K E^j$; its binding layer w_{K-1} has length $1 - \frac{3}{4}\cdot 2^{-j}$ relative to its own scale, **independent of** K, so the route needs $j \ge 1$. This rules out $OEOEOE$ ($j = 0$, relative length $\frac{1}{4}$, where the phase is essentially linear and the measured worst-case cancellation is nil) and admits the family $V_k = (OE)^{k-1}OEE$ — OEE, $OEOEE$, $OEOEOEE$, … — every member at relative length $\frac{5}{8}$, with ideal coefficient $c_k = 3^{-k}$ and net gain $\frac{1}{9}c_k = 3^{-(k+2)}$ at scale $(3/4)^k\frac{3}{8}$. With $x = 2^{-\lambda}$, $y = (3/4)^\lambda$ the family telescopes: $x + \frac{1}{9}xy + \frac{2}{9}y + \sum_{k\ge1} 3^{-(k+2)}xy^{k+1} = 1 \iff x + \frac{1}{3}y = 1$, since the family sum is $\frac{1}{27}xy^2/(1-\frac{1}{3}y)$, which at $x = 1-\frac{y}{3}$ is $\frac{1}{27}y^2$, exactly cancelling the $-\frac{1}{27}y^2$ from $\frac{1}{9}xy$. The right-hand equation is the ideal depth-two recursion, root 0.4927: **the elementary family closes the whole depth-two gap**, with finite truncations 0.4801, 0.4891, 0.4916, 0.4924, and 0.5769 alongside Appendix C.

V_2, V_3 and V_4 are worked out in full, and the pattern closes. For V_k the layers are $n, w_1, \dots, w_{k-1}$, with w_i at scale $P^{(3/4)^i}$ and block length $\frac{1}{3}(3/4)^i$; the binding layer is w_{k-1}, **always at relative length** $\frac{5}{8}$. The sign sums fall into $k-1$ Cauchy–Schwarz cases — the one whose deepest varying factor sits at layer $i-1$ is summed over layer i, with the Vaaler truncation balanced at $3S = H_i$ and saving $H_i/3$ — plus pairing cases and one one-variable case at layer $k-1$. Summing $|S_q|$ directly rather than through Cauchy–Schwarz, and balancing every Vaaler truncation, gives $H_i/2$ for the Cauchy–Schwarz cases and $\frac{1}{6}(3/4)^{k-1}$ for the one-variable case; these coincide, since $H_{k-1}/2 = \frac{1}{2}\cdot\frac{1}{3}(3/4)^{k-1}$, so

$$\text{binding saving for } V_k = \tfrac{1}{6}\left(\tfrac{3}{4}\right)^{k-1} = \tfrac{1}{8},\ \tfrac{3}{32},\ \tfrac{9}{128},\ \tfrac{27}{512}, \dots$$

geometrically decaying but never zero: **every V_k carries a power saving, so every truncation of the family is a theorem.** The net gains are $\frac{1}{9}c_k = 3^{-(k+2)}$ at ρ_{k+1}, giving 0.4801, 0.4891, 0.4916 and, alongside Appendix C, 0.5665, 0.5740, 0.5761.

The censuses confirm the exact identities and the block structure without a single failure at any layer, and every conditional share a single fiber can sample reads 0.50. They do **not** confirm the constants 2^{-2k} for $k \ge 3$: the deepest layers of V_k are sampled at only P^{ρ_k}-ish many points — 50 and 7 distinct values at $P = 10^8$ for V_4 — so those shares are noise, and reaching 100 samples at V_4's deepest layer needs $P \approx 10^{12.7}$. The reduction, the telescoping identity and the census are complete; the explicit constants in the two half-estimates are not computed here, so this is recorded as a family in reserve rather than as a replacement for Theorem 1's exponent (juggler_oeoee_production.md).

6. Odd generation and the exact first-letter decomposition

Write $S = \{\lfloor m^{3/2}\rfloor : m \text{ odd}\}$ for the image of the odd integers, and S_{odd} for its odd members. The set S is sparse: it has $\asymp z^{2/3}$ members below z.

6.1 Odd generation

Theorem 6.1 (odd generation; Lean). Let A be forward- and backward-closed with $1 \notin A$. Then:

1. every $n \in A$ descends by even steps only to an odd member $m \ge 3$ of A (`exists_odd_ancestor_ge_three`);
2. for odd n, $n \in A \iff \lfloor n^{3/2}\rfloor \in A$ (`odd_mem_iff`), so the odd members of A are exactly the odd preimages of $A \cap S$;
3. $A \ne \emptyset \iff A \cap S \ne \emptyset$, i.e. A contains the image of some odd $m \ge 3$ (`nonempty_iff_odd_image_mem`).

Proof. (1) If $n \in A$ is even, $\lfloor\sqrt{n}\rfloor \in A$ by forward closure and is smaller; the chain stops at an odd integer, which is ≥ 3 since $1 \notin A$ and $2 \notin A$ ($J(2) = 1$). (2) Forward closure gives one direction, backward closure the other. (3) follows from (1) and (2). □

Applied to the failure set F: every positive integer reaches 1 iff no image $\lfloor m^{3/2}\rfloor$ of an odd m fails to reach 1, and F is the E-forest over the odd preimages of $F \cap S$. The theorem changes the geometry of the problem. An arbitrary failure set is replaced by a set of *seeds* — odd images, a set of density $\asymp z^{-1/3}$ at scale z — together with the deterministic even ancestry of each seed (its E-forest, whose levels are the intervals of Lemma 3.1) and the unique odd preimage of each odd seed. Two consequences are used below: the log-count of F is controlled by that of its odd members (Theorem 7.2), and the first-letter decomposition of F has exactly one term that the productions do not control (Section 6.2).

6.2 The exact decomposition and the free term

Let A be forward- and backward-closed. On $(\sqrt{x}, x]$ its members split by first letters into three exact pieces (Figure 3):

- the **even** members, $= \bigsqcup_{m\in A} E(m) \cap (\sqrt{x}, x]$, of log-mass $\ell_A(x^{1/4}, \sqrt{x}] + O(x^{-1/4})$ (Lemma 3.1);
- the OE**-type** odd members ($\lfloor n^{3/2}\rfloor$ even), $= \bigsqcup_{m\in A}\{n \in \Phi(m) : \lfloor n^{3/2}\rfloor \text{ even}\}$ (Lemma 3.2), of log-mass between $\frac{2}{9}$ and $\frac{4}{9}$ times $\ell_A(x^{3/8}, x^{3/4}]$ on good fibers (Lemma 4.2, two-sided pairing) and $\frac{1}{3}(1 + o(1))$ times it on E-blocks (Proposition 4.4);
- the OO**-type** odd members ($\lfloor n^{3/2}\rfloor$ odd), = the odd preimages of $A \cap S_{\text{odd}} \cap (x^{3/4}, x^{3/2}]$, of log-mass $\sum_{m\in A\cap S_{\text{odd}}} 1/n(m)$, where $n(m)$ is the unique odd preimage.

Let $\varphi_A(t) = \ell_A(e^{t/2}, e^t]/(t/2)$ be the logarithmic density of A on $(\sqrt{x}, x]$, $t = \log x$. Define $\varphi_A^{\text{fib}}(3t/4)$ by the normalization $\frac{1}{4} \cdot \frac{t}{2} \cdot \varphi_A^{\text{fib}}(3t/4) =$ log-mass of the OE-type members, and $\psi_A(t)$ by $\frac{1}{4} \cdot \frac{t}{2} \cdot \psi_A(t) =$ log-mass of the OO-type members. Then $\varphi_A^{\text{fib}}(s)$ is the density of A on $(e^{s/2}, e^s]$ weighted, on each fiber, by twice the even-image fraction $p_m = G_m/H_m$ of that fiber — so $\varphi_A^{\text{fib}}(s) = \varphi_A(s)(1 + o(1))$ when A consists of E-blocks on that range (Proposition 4.4), and $\varphi_A^{\text{fib}}(s) \leq \frac{8}{7}\varphi_A(s) + O(e^{-s/6})$ always (Lemmas 4.2, 4.3) — and $\psi_A(t)$ is the log-weighted share of A among the OO-type odd integers of $(\sqrt{x}, x]$, up to a factor $1 + O(e^{-\delta t})$ from the depth-two equidistribution of the parity of $\lfloor n^{3/2}\rfloor$. With these definitions the three pieces give the exact identity

$$\varphi_A(t) = \tfrac{1}{2}\varphi_A(t/2) + \tfrac{1}{4}\,\varphi_A^{\text{fib}}(3t/4) + \tfrac{1}{4}\,\psi_A(t) + O(e^{-t/4}/t), \tag{6.1}$$

whose only error is the boundary term of Lemma 3.1. For $A = \mathbb{N}$ every term is $1 + o(1)$. For $A = F$ the boundary condition is $\varphi_F(t) = 0$ for $t \leq \log N_0$.

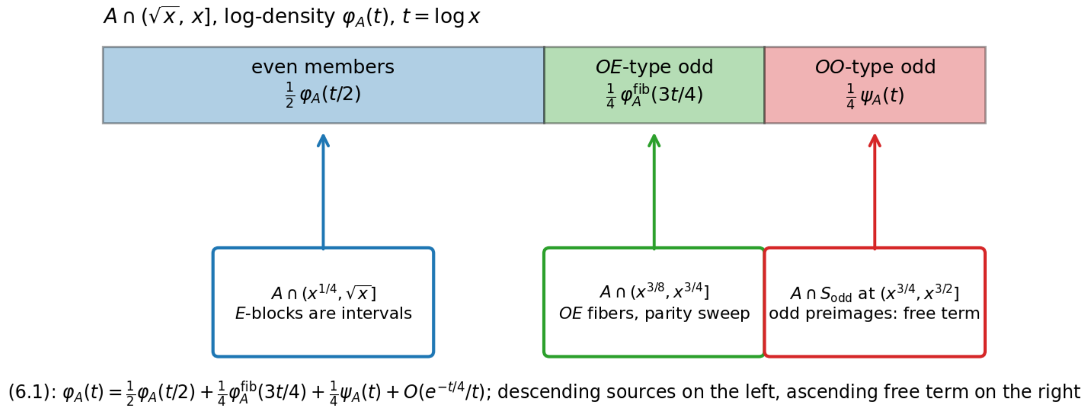

Figure 3: The first-letter decomposition of a two-way closed set on $(\sqrt{x}, x]$: the even members come from scale $x^{1/2}$ through E-blocks, the OE-type odd members from scale $x^{3/4}$ through fibers, and the OO-type odd members are the odd preimages of the odd images of A at scale $x^{3/2}$ — the free term.

Definition 6.2 (S-fairness). A two-way closed set A is *S-fair at scale t* if $\psi_A(t) \le \varphi_A(3t/2)$, i.e. if A is not over-represented among the odd images of odd integers at scale $e^{3t/2}$ relative to its density among all integers there; it is *S-fair* if this holds for all $t \ge t_0$.

Proposition 6.3 (exact statements). (i) $\psi_F \equiv 0$ iff $F = \emptyset$: if $F \ne \emptyset$, its minimum is odd with odd image, so $\psi_F(t) > 0$ at the scale of $\min F$. (ii) The identity (6.1) holds for F with the stated error, and its right side contains exactly one term, $\frac{1}{4}\psi_F(t)$, that is not determined by F at smaller scales through the two productions.

Proof. (i) If $F = \emptyset$ then $\psi_F \equiv 0$. If $F \ne \emptyset$, let $n_F = \min F$. It is odd (an even failure has the smaller failure $\lfloor\sqrt{n_F}\rfloor$), and its image is odd (if $J(n_F)$ were even, $J^2(n_F) \approx n_F^{3/4} < n_F$ would be a smaller failure). So n_F is an OO-type failure and $\psi_F(t) > 0$ for t with $e^{t/2} < n_F \le e^t$. (ii) is the decomposition. □

Remark 6.4 (the walk heuristic; not a theorem). If F were S-fair, if the fiber weights were ideal ($\varphi_F^{\mathrm{fib}} = \varphi_F$), and if the error term of (6.1) were zero, then (6.1) would give $\varphi_F(t) \le \frac{1}{2}\varphi_F(t/2) + \frac{1}{4}\varphi_F(3t/4) + \frac{1}{4}\varphi_F(3t/2)$: the harmonic inequality of a random walk on $\log t$ with steps $\log\frac{1}{2}, \log\frac{3}{4}, \log\frac{3}{2}$ of probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$, whose drift -0.317 is negative, so that the boundary condition $\varphi_F = 0$ below $\log N_0$ would force $\varphi_F \equiv 0$. None of the three conditions is available with the constants proved here: the fiber weights are only known to lie in $[\frac{2}{3}, \frac{4}{3}]$, so the coefficients of the inequality can sum to $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12} > 1$, and the boundary error $O(e^{-t/4}/t)$ is not zero at the scales just above the floor where the walk exits. We therefore do not claim that S-fairness implies the conjecture. What is exact is Proposition 6.3; what is proved about the free term is Proposition 10.1, which identifies $\psi_F(t)$ with a quantity bounded by every hypothesis of Sections 8–9.

6.3 Where Papers A and B sit

Paper A [11] constrains the fate Lachesis from the inside: finance and the walk charge bound the *states* of a hypothetical cycle (minimum $> 3.5 \cdot 10^8$, period ≥ 780239). Theorem 1 constrains the basin: if the cycle exists, its basin is a two-way closed class with the cycle's states as seeds and log-count $\gg (\log x)^{0.448}$. The two do not meet: finance bounds the seed, contagion the growth from the seed, and no inequality bounds a basin from above. Paper B [12] controls the descending branches of (6.1) — the fairness of the landing distributions of E, OE and, with its depth-4 and depth-5 theorems, of deeper contracting words — on dyadic blocks; Appendix C uses one statement of that type on sub-dyadic intervals as an explicit hypothesis. Neither touches ψ_F: the ascending branch OO sends mass to $x^{3/2}$, and its return is the nested-floor parity at all depths.

7. The almost-all equivalence

Corollary 7.1 (logarithmic form). The following are equivalent: (1) every $n \geq 1$ reaches 1; (2) for some $\lambda < \lambda^{**}$, $\sum_{n \leq x,\ n \notin R} 1/n = o((\log x)^{\lambda})$; (3) for some $\lambda < \lambda^{**}$, the starts $n \leq x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^{\lambda})$.

Proof. (1)⇒(3): the set is empty. (3)⇒(2): an orbit that enters $[1, N_0]$ reaches 1, so F is contained in the set of (3). (2)⇒(1): F is backward-closed; if it were nonempty, Theorem 5.3 would contradict (2). □

Theorem 7.2 (a Tao-type bound with rate implies the conjecture). Suppose that for some $e > 1 - \lambda^{**} = 0.5520\ldots$ and all sufficiently large y,

$$\#\{n \text{ odd},\ y < n \leq 2y :\ n \notin R\} \ \leq\ \frac{y}{(\log y)^{e}}.$$

Then $R = \mathbb{N}$.

Proof. Suppose $F \neq \emptyset$. By Theorem 6.1 every $n \in F$ lies in the E-tree of an odd member of F. For an odd $n_0 \in F$ the level-j set $S_j(n_0)$ of its E-tree has log-mass at most $\prod_{i<j}(1 + n_0^{-2^i})/n_0 \leq 2/n_0$ (Lemma 3.1 gives $\sum_{n \in E(m)} 1/n \leq (m+1)/m^2$), and only levels with $n_0^{2^j} \leq x$, i.e. $j \leq \log_2 \log x$, meet $[1, x]$. Hence

$$\sum_{\substack{n \in F \\ n \leq x}} \frac{1}{n} \ \leq\ 2\,(1 + \log_2 \log x) \sum_{\substack{n_0 \in F \text{ odd} \\ n_0 \leq x}} \frac{1}{n_0} \ \leq\ 2\,(1 + \log_2 \log x)\Big(\sum_{k \geq k_0} \frac{2^k (k \ln 2)^{-e}}{2^k} + O(1)\Big) \ \ll\ (\log x)^{1-e} \log\log x.$$

By Theorem 5.3 the left side is $\geq K(\log x)^{\lambda}$ for every $\lambda < \lambda^{**}$ and all large x. Choosing $\lambda \in (1 - e, \lambda^{**})$ gives a contradiction. □

Theorem 7.3 (equivalence). Every positive integer reaches 1 if and only if there is $e > 1 - \lambda^{**}$ such that $\#\{n \text{ odd} \in (y, 2y] :\ n \notin R\} \leq y(\log y)^{-e}$ for all large y.

Proof. If every integer reaches 1 the left side is 0. Conversely Theorem 7.2. □

The threshold $1 - \lambda^{**}$ is the complement of the contagion exponent; with λ^{***} of Appendix C it becomes 0.4608. Any improvement of the contagion exponent lowers the rate required of the almost-all statement, and $\lambda \to 1$ would make any positive rate suffice — but that improvement requires all descent certificates and is circular here (Section 5.6).

7.2 Comparison with the Collatz map: a structural analogy

The comparison with Tao's theorem [6] is a structural analogy, not a transfer of methods, and it concerns two specific features.

Depth. Juggler descent is by powers — one contracting certificate sends n to n^e, $e < 1$ — so the exponent walk of an itinerary has to travel $\log_2 \log y$ units to bring a start of size y below a fixed N_0, and the parity word needed is $O(\log\log y)$ letters long, a depth at which itinerary cylinders still contain $y/2^d \gg 1$ starts. Collatz descent is by constant factors, the walk needs $\Theta(\log y)$ steps, and cylinders of that depth have single elements; a growing target $f(N) \to \infty$ and the renewal structure of the Syracuse random variables are what make Tao's argument possible.

Preimage trees. For the Juggler map, Theorem 1 gives every preimage tree logarithmic count $\gg (\log x)^{\lambda}$, which a bound of the form $y(\log y)^{-e}$, $e > 1 - \lambda$, on failures contradicts. For the Collatz map the known lower bound for a preimage tree is $x^{0.84}$ below x [7]; a set of polynomial size $x^{0.84}$ can have logarithmic count tending to zero, so a logarithmic error rate on failures is compatible with a nonempty failure tree of that size, and the passage from almost all to all is not available from this information. The distinction is one of growth scales — polynomial against logarithmic — not a dichotomy between "thin" and "fat" trees.

What is exact for Collatz — the equidistribution of parity vectors mod 2^k, Terras [4], Everett [5] — is the unproved hypothesis of Section 8; what is difficult for Collatz — renewal to a bounded target and the passage from almost all to all — is supplied here by the power descent and by Theorem 1. Other ingredients of Tao's proof have no counterpart in this paper and are not claimed to.

8. From parity control to the almost-all bound

8.1 Notation

For a word $w \in \{O,E\}^d$ and $t \le d$ write $o_t(w)$ for the number of odd letters among the first t and

$$u_t(w) = o_t(w)\log_2 3 - t$$

for the *exponent walk*, so that the ideal image after t letters is $n^{2^{u_t}}$. For $y \ge 1$ put

$$L(y) = \log_2 \frac{\log 2y}{\log N_0}, \qquad d(y) = \lceil C\,L(y)\rceil,$$

with a constant $C \ge 5$ fixed below. A word w of length d is *L-bad* if $u_t(w) > -L$ for every $1 \le t \le d$ (its walk never descends to $-L$). $\mathrm{word}_d(n)$ is the realized itinerary of the first d steps of n (letter s is the parity of $J^{s-1}(n)$); the *cylinder* $[w]_y$ is the set of odd $n \in (y,2y]$ with $\mathrm{word}_{|w|}(n) = w$. Odd starts have first letter O, so the fair share of an O-rooted cylinder of depth d among the $N = y/2$ odd starts of $(y,2y]$ is $2^{-(d-1)}N$.

8.2 Envelope descent and the rarity of bad words

Lemma 8.1 (envelope descent). Let $n \in (y,2y]$ and $d \ge 1$. If $\mathrm{word}_d(n)$ is not $L(y)$-bad, then $n \in R$.

Proof. Some $t \le d$ has $u_t \le -L(y)$, i.e. $3^{o_t}/2^t \le \log N_0/\log 2y \le \log N_0/\log n$, i.e. $n^{3^{o_t}} \le N_0^{2^t}$ as integers. The power envelope $J^t(n)^{2^t} \le n^{3^{o_t}}$ (Paper A, Theorem 2.2; `power_bound_word`) gives $J^t(n) \le N_0$, so $J^t(n) \in R$ and $n \in R$ by backward closure. Lean: `iterate_le_of_envelope`, `mem_of_envelope_floor`, `reachesOne_of_itinerary_envelope`. ☐

Lemma 8.2 (Chernoff). For $C \ge 5$ put

$$p_C = \frac{1-1/C}{\log_2 3}, \qquad e(C) = \frac{C\,D(p_C \,\|\, \frac12)}{\ln 2}, \qquad D(p\|\tfrac12) = p\ln(2p) + (1-p)\ln(2(1-p)).$$

For $L > 0$ and $d \ge CL$, the number of L-bad words of length d is at most $2^d \cdot 2^{-e(C)L}$. Moreover, for every $\varepsilon > 0$ there is L_0 such that for $L \ge L_0$ and $d = \lceil CL\rceil$ the number of L-bad words of length d beginning with O is at most $2^{d-1}\cdot 2^{-(e(C)-\varepsilon)L}$.

Proof. An L-bad word has $u_d > -L$, i.e. $o_d > (d-L)/\log_2 3 \ge p_C d$ (as $L/d \le 1/C$). For a fair coin, $\Pr[\mathrm{Bin}(d,\frac12) \ge pd] \le e^{-dD(p\|1/2)}$ for $p \ge \frac12$, and $D(\cdot\|\frac12)$ is increasing on $[\frac12,1]$; $p_C \ge \frac12$ for $C \ge 5$. Hence the count is at most $2^d e^{-dD(p_C\|1/2)} \le 2^d e^{-CL\,D(p_C\|1/2)} = 2^d 2^{-e(C)L}$. For the O-rooted count, the remaining $d-1$ letters contain $o_d - 1 > (d-L)/\log_2 3 - 1 = p'(d-1)$ odd letters with $p' = p_C - O(1/L)$; the same bound with $d-1$ trials gives $2^{d-1}e^{-(d-1)D(p'\|1/2)}$, and $(d-1)D(p'\|1/2) \ge (e(C)-\varepsilon)L\ln 2$ for L large. ☐

Numerically $e(18) = 0.480$, $e(19) = 0.527$, $e(20) = 0.574$, $e(21) = 0.621$, $e(25) = 0.812$, $e(30) = 1.054$; $e(C)$ grows linearly in C with slope $D(1/\log_2 3 \,\|\, \frac12)/\ln 2 = 0.050$.

8.3 The cylinder hypothesis and the bound

Hypothesis $\mathrm{H}(C,A)$ **(log-log-depth cylinder bound).** For all sufficiently large y, with $d = d(y)$, every word $w \in \{O,E\}^d$ beginning with O satisfies

$$\#[w]_y \;\le\; 2^{-(d-1)}\cdot\frac{y}{2} + \frac{y}{(\log y)^A}.$$

Only an upper bound is asked, only at one depth per scale, and only for the $L(y)$-bad words. Since $2^{-(d-1)}y/2 \asymp y(\log y)^{-C}$, $\mathrm{H}(C,A)$ with $A > C$ says that no O-rooted cylinder of depth $d(y)$ exceeds its fair share by more than a relative $O((\log y)^{C-A})$.

Theorem 8.3 (Tao-type bound from the cylinder hypothesis). Assume $\mathrm{H}(C,A)$ with $C \geq 5$ and $A > C + e(C)$. Then for every $\varepsilon > 0$ and all sufficiently large y,

$$\#\{n \text{ odd} \in (y,2y] : \ n \notin R\} \ \leq \ \#\{n \text{ odd} \in (y,2y] : \ J^t(n) > N_0 \ \forall t \leq d(y)\} \ \leq \ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e(C)-\varepsilon)}.$$

In words: all but $O(y(\log y)^{-e(C)+\varepsilon})$ odd starts in $(y,2y]$ enter $[1,N_0]$ within $d(y) = O(\log\log y)$ steps.

Proof. The first inequality is Lemma 8.1 with $d = d(y)$. Every odd start has an O-rooted word; by Lemma 8.2 the O-rooted bad words number at most $2^{d-1}2^{-(e(C)-\varepsilon)L(y)}$, and by $\mathrm{H}(C,A)$ each carries at most $2^{-(d-1)}y/2 + y(\log y)^{-A}$ odd starts. Hence the count is at most

$$\frac{y}{2}\,2^{-(e(C)-\varepsilon)L(y)} + 2^{d-1}\,\frac{y}{(\log y)^A} \ \leq \ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e(C)-\varepsilon)} + \Big(\frac{\log 2y}{\log N_0}\Big)^{C}\frac{y}{(\log y)^A},$$

using $2^{d-1} \leq 2^{CL}$. The second term is $O(y(\log y)^{C-A}) = o(y(\log y)^{-e(C)})$ because $A > C + e(C)$. ☐

Corollary 8.4 (the conjecture from a cylinder bound). If $\mathrm{H}(C,A)$ holds for some $C \geq 20$ and $A > C + e(C)$, then every positive integer reaches 1. Under the conditional exponent λ^{***} of Appendix C the same conclusion holds for $C \geq 18$.

Proof. Theorem 8.3 gives the hypothesis of Theorem 7.2 with $e = e(C) - \varepsilon \geq e(20) - \varepsilon = 0.574 - \varepsilon > 0.5520$; with λ^{***} the threshold is $0.4608 < e(18) = 0.480$. ☐

8.4 Constants

With the certified floor $N_0 = 3.5 \cdot 10^8$ and the least unconditional depth $C = 20$:

y	$L(y)$	$d(y)$	exact fair-coin bad probability	$(\log y)^{-0.6}$	least depth for rate 0.6
10^{20}	1.25	25	0.065	0.100	19
10^{100}	3.55	72	0.017	0.038	56
10^{1000}	6.87	138	0.0038	0.0096	117

The bad probability is the fair-coin probability that a walk of length $d(y)$ never reaches $-L(y)$, by dynamic programming over the walk (unconditioned first letter; conditioning on $u_1 = \log_2 3 - 1$ roughly doubles the finite-depth values and leaves the exponent unchanged); the last column is the least depth at which it drops below $(\log y)^{-0.6}$ — about $17\,L(y)$. With the Lean floor $N_0 = 260$ the depths grow by about 38 letters. For comparison, Paper B controls depth 4 (all words) and two words of depth 5, with relative error $y^{-1/96}$ on dyadic blocks; the hypothesis asks for depth $d(y) \to \infty$ with relative error $o(1)$ — weaker in rate than a power saving, unbounded in depth.

9. Weaker forms of the hypothesis

Theorem 8.3 conditions on the past letter by letter and asks for two-sided control of every cylinder. Neither is needed. This section gives the hierarchy of forms down to the weakest one the concentration argument can use, and records what that form does not need.

9.1 One-sided odd-share control

Hypothesis $\mathrm{H}_q(C,A)$**.** For all sufficiently large y, every $1 \leq t < d(y)$ and every $w \in \{O,E\}^t$,

$$\#\{n \in [w]_y : \ \mathrm{word}_{t+1}(n) = wO\} \ \leq \ q\,\#[w]_y + \frac{y}{(\log y)^A}.$$

(The depth-0 cylinder is all odd starts, whose next letter is O with share 1; the hypothesis starts at depth 1.)

Theorem 9.1 (one-sided form). Let $q < \log 2/\log 3 = 0.6309\ldots$, $\mu = 1 - q\log_2 3 > 0$, $C > 1/\mu$, and $e_q(C) = 2(C\mu - 1)^2/(C(\log_2 3)^2 \ln 2)$. Under $\mathrm{H}_q(C, A)$ with $A > C + 1$, for every $\varepsilon > 0$ and all large y,

$$\#\{n \text{ odd} \in (y, 2y] :\ n \notin R\} \ \le\ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e_q(C)-\varepsilon)}.$$

Hence $e_q(C) > 1 - \lambda^{**}$ implies the conjecture; the least C is 20 at $q = \frac{1}{2}$, 44 at 0.55, 240 at 0.60, 1715 at 0.62 (with λ^{***}: 18, 39, 206, 1451).

Proof. Let n be uniform on the odd integers of $(y, 2y]$, $\mathcal{F}_t$ the σ-algebra of the depth-t cylinders, $X_t = \mathbf{1}[\mathrm{word}_{t+1}(n) = \mathrm{word}_t(n)O]$ for $t \ge 1$, and $u_t = (\log_2 3 - 1) + \sum_{1\le s<t}(X_s \log_2 3 - 1)$ the exponent walk. Put $\eta_t = (\mathbb{P}(X_t = 1 \mid \mathcal{F}_t) - q)^+$. On a cylinder $[w]$ of depth t, H_q gives $\eta_t \le y(\log y)^{-A}/\#[w]$, so $\mathbb{E}[\eta_t] \le 2^{t+1}(\log y)^{-A}$ and $\mathbb{E}[\sum_{1\le t<d}\eta_t] \le 4(\log 2y/\log N_0)^C(\log y)^{-A}$. Since $\mathbb{E}[u_{t+1} - u_t \mid \mathcal{F}_t] \le -\mu + \eta_t \log_2 3$, the process $M_t = u_t - u_1 - \sum_{1\le s<t}\mathbb{E}[u_{s+1} - u_s \mid \mathcal{F}_s]$ is a martingale with increments in an interval of length $\log_2 3$, and $u_d \le u_1 + M_d - (d-1)\mu + \log_2 3\sum_{s<d}\eta_s$. Fix $\kappa \in (0, 1)$. If $n \notin R$ then by Lemma 8.1 $u_d > -L$, so either $\log_2 3\sum_s \eta_s > \kappa(d-1)\mu$, an event of probability $O((\log y)^{C-A})$ by Markov's inequality, or $M_d > a := (1-\kappa)(d-1)\mu - L - u_1 \ge L((1-\kappa)C\mu - 1) - \log_2 3$. By the Azuma–Hoeffding inequality, $\mathbb{P}(M_d > a) \le \exp(-2a^2/((d-1)(\log_2 3)^2)) \le 2^{-L(e_q^{(\kappa)}(C)-o(1))}$ with $e_q^{(\kappa)}(C) = 2((1-\kappa)C\mu - 1)^2/(C(\log_2 3)^2 \ln 2) \to e_q(C)$ as $\kappa \to 0$. $\square$

So no lower bound on odd shares and no vanishing error are needed: if no cylinder of depth below $44\log_2(\log 2y/\log N_0)$ sends more than 55% of its members to an odd state, every positive integer reaches 1.

9.2 The pressure form

Let $N = y/2$, let $\tau(n) = \min\{t \ge 1 :\ J^t(n) \le N_0\}$ be the entrance time into the floor, call n *live at depth* t if $\tau(n) > t$, and write $o_t(n)$ for the number of odd letters among $n, J(n), \ldots, J^{t-1}(n)$. By Lemma 8.1, $\tau(n) > t$ implies $u_t(n) > -L(y)$. The stopping is essential: $J(1) = 1$ is odd, so an orbit that has reached 1 has an all-O tail and the unstopped odd count of every start in R grows linearly. For $\theta > 0$ put $a_\theta = \frac{1}{2}(1 + e^\theta)$.

Hypothesis $\mathrm{P}_\theta(C)$ (live pressure). For all sufficiently large y, with $d = d(y)$,

$$\frac{1}{N}\sum_{\substack{n \ \mathrm{odd}\in(y,2y] \\ \tau(n)>d}} e^{\theta\, o_d(n)} \ \le\ a_\theta^d\, e^{o(d)}.$$

Theorem 9.2 (pressure form). Assume $\mathrm{P}_\theta(C)$ with $C \ge 5$ and $\theta = \theta_C = \log(p_C/(1-p_C))$. Then for every $\varepsilon > 0$ and all large y,

$$\#\{n \text{ odd} \in (y, 2y] :\ \tau(n) > d(y)\} \ \le\ \frac{y}{2}\Big(\frac{\log 2y}{\log N_0}\Big)^{-(e(C)-\varepsilon)},$$

the bound of Theorem 8.3. Hence $\mathrm{P}_{\theta_C}(C)$ with $C \ge 20$ implies the conjecture unconditionally, and with $C \ge 18$ under the conditional exponent of Appendix C.

Proof. If $\tau(n) > d$ then $u_d > -L$, i.e. $o_d \ge p_C d$. Hence

$$\#\{\tau > d\} \le \#\{\tau > d,\ o_d \ge p_C d\} \le e^{-\theta p_C d}\sum_{\tau(n)>d} e^{\theta o_d(n)} \le N\, e^{-\theta p_C d} a_\theta^d e^{o(d)}.$$

At $\theta = \theta_C$, $\theta p_C - \log a_\theta = D(p_C\|\frac{1}{2})$, so the count is $\le N\exp(-dD(p_C\|\frac{1}{2})(1 - o(1))) \le N2^{-(e(C)-\varepsilon)L}$. $\square$

For $1 \le t < d$ let $\mu_{\theta,t}$ be the probability measure on the starts live at depth t with density proportional to $e^{\theta o_t(n)}$, and let $s_\theta(t) = \mu_{\theta,t}(J^t(n) \text{ odd})$ be the *tilted odd share*: the probability that the next letter is O when odd-heavy live prefixes are up-weighted exponentially.

Hypothesis $\mathrm{M}_{\theta,q}(C)$ (no momentum). For all large y, $\sum_{t=1}^{d(y)-1}(s_\theta(t) - q)^+ = o(d(y))$.

Proposition 9.3. $\mathrm{M}_{\theta,1/2}(C)$ implies $\mathrm{P}_\theta(C)$. More generally $\mathrm{M}_{\theta,q}(C)$ with $q < p_C$ gives $\#\{\tau > d\} \leq N\exp(-d\,D(p_C\|q)(1-o(1)))$, i.e. the exponent $e_q^{\mathrm{Ch}}(C) = C\,D(p_C\|q)/\ln 2$, at least the Azuma exponent of Theorem 9.1 (least C: $20, 43, 230, 1618$ at $q = 0.5, 0.55, 0.60, 0.62$; with λ^{***}: $18, 38, 198, 1369$).

Proof. Dropping the condition $\tau > t+1$ in favour of $\tau > t$ only enlarges the sum, so

$$\sum_{\tau>t+1} e^{\theta o_{t+1}} \leq \sum_{\tau>t} e^{\theta o_{t+1}} = \Big(\sum_{\tau>t} e^{\theta o_t}\Big)(1+(e^\theta-1)s_\theta(t)).$$

With $a_{\theta,q} = 1+(e^\theta-1)q$ and $c_\theta = (e^\theta-1)/a_{\theta,q}$, $1+(e^\theta-1)s \leq a_{\theta,q}\exp(c_\theta(s-q)^+)$. Telescoping from $t=1$ gives $\sum_{\tau>d} e^{\theta o_d} \leq Ne^\theta a_{\theta,q}^{d-1}\exp(c_\theta\sum_{t<d}(s_\theta(t)-q)^+)$, which is P_θ at $q = \frac{1}{2}$; the exponential Markov step of Theorem 9.2 with $a_{\theta,q}$ in place of a_θ and $\theta = \log\frac{p_C(1-q)}{q(1-p_C)}$ gives $D(p_C\|q)$. $\square$

9.3 What the weakest form does not need

(a) Any $o(\log\log y)$ initial depths are free. A letter contributes a factor at most e^θ to the moment, so bias — however extreme — at any set of depths of size $o(d(y))$ is absorbed by the $e^{o(d)}$. In particular the depth-five split (Paper B's Conjecture 7.3) and every split to any fixed depth, or to depth $s_0(y)$ for any $s_0 = o(\log\log y)$, is irrelevant to the reduction.

(b) The tower cylinders are nearly free. A cylinder w of depth t enters P_θ only through $\#[w]e^{\theta o(w)}$, against a fair total $Ne^\theta a_\theta^{t-1}$. If the tower O^t splits with odd share β at every level, $\#[O^t] = N\beta^{t-1}$ and its tilted weight relative to the fair total is $(\beta e^\theta/a_\theta)^{t-1}$, which decays geometrically iff $\beta < \frac{1}{2}(1+e^{-\theta})$, i.e. 0.836 at $\theta_{19} = 0.396$. So a tower biased anywhere below 0.836 contributes a bounded total to $\sum_t(s_\theta(t)-\frac{1}{2})^+$ when the rest of the population is fair; H_q needs the same tower to split below 0.6309, and the conclusion of Theorem 8.3 itself fails only once $\beta > 2^{-e(C)/C} = 0.981$. A single cylinder of depth t matters only if it is over-populated by a factor $(2a_\theta e^{-\theta})^t = (1.67)^t$ relative to its fair share.

(c) One-sidedness and aggregation. $\mathrm{M}_{\theta,q}$ bounds one weighted average per depth from above; under-populated odd continuations anywhere, and over-populated ones on cylinders of small tilted weight, cost nothing. In Walsh terms, $e^{\theta X_s} = a_\theta + b_\theta(-1)^{J^s(n)}$ with $b_\theta/a_\theta = -\tanh(\theta/2)$, so the unstopped moment is

$$\frac{1}{N}\sum_n e^{\theta o_d(n)} = a_\theta^d\Big(1+\sum_{\emptyset\neq T}(-\tanh\tfrac{\theta}{2})^{|T|}\frac{W_T}{N}\Big), \qquad W_T = \sum_n\prod_{s\in T}(-1)^{J^s(n)}:$$

Walsh sums of high order are exponentially down-weighted ($\tanh(0.198) = 0.196$ per letter), whereas the uniform cylinder hypothesis needs all 2^d of them with equal weight.

(d) Intermediate forms are reparameterizations. Between $\mathrm{H}(C,A)$ and P_θ sit the "almost all cylinders" form (odd share $\leq q$ on all but a family of cylinders of total mass $\leq N(\log y)^{-B}$ per depth, $B > e_q(C)$; the Markov step of Theorem 9.1 absorbs it) and its second-moment form. With $D(w) = \#[wO] - \frac{1}{2}\#[w]$ and the collision count $\mathcal{C}_t = \sum_{|w|=t}\#[w]^2$, Parseval's identity gives $\sum_w D(w)^2 = 2^{-t-2}\sum_{S\subseteq[0,t)}|W_{S\cup\{t\}}|^2 = \frac{1}{2}\mathcal{C}_{t+1} - \frac{1}{4}\mathcal{C}_t$, so the almost-all form follows from the pair-correlation asymptotic $\mathcal{C}_t \leq (N^2/2^{t-1})(1+(\log y)^{-A'})$, $A' > C+2$; but by Walsh inversion any such two-sided bound is equivalent, up to the exponent, to polylogarithmic savings on every Walsh sum to depth $d(y)$, i.e. to $\mathrm{H}(C,A)$ with another A. They are not weaker in substance; the pressure form is.

(e) No bounded-depth cylinder statement implies the bound. Fix k. The probability measure on parity words that is fair to depth k and all-O afterwards satisfies every cylinder-count statement of depth $\leq k$ exactly, and under it a positive proportion of starts is live for ever. An argument for the Tao-type bound that used only depth-$\leq k$ cylinder statistics of the Juggler map and the walk logic would prove the same bound for that measure. Hence every proof of the bound — in any of the forms above — must use information about the parity words at depth $\to\infty$; fixed-depth equidistribution at any depth is neither necessary (by (a)) nor sufficient (by (e)) for it.

9.4 Where the difficulty lies

$\mathrm{P}_\theta(C)$ says that the exponentially tilted distribution of live parity words of length $d \asymp \log_2 \log y$ has the fair pressure $\log a_\theta$: odd-heavy live words — those with about $e^\theta/(1+e^\theta) \approx 60\%$ odd letters, the words the tilt selects — are not over-populated at an exponential rate in d. Typical such words contain odd runs of length ≥ 4, so the nested towers of height ≥ 4 that stop Paper B at depth five sit inside almost every relevant cylinder; but no individual cylinder is required to split fairly, only the tilted average over the $\approx 2^{0.97d}$ typical cylinders at each depth. This is a statement about averaging over cylinders at unbounded depth — a mean over characters, not a supremum — of a kind for which analytic number theory sometimes has tools where pointwise bounds fail. We record the question and do not pursue an analytic approach here.

10. One frontier statement

10.1 The free term is the infinite-depth live mass

Recall the free term ψ_F of (6.1). For odd n, $n \in F \iff \lfloor n^{3/2} \rfloor \in F$ (Theorem 6.1), and $n \in F \iff \tau(n) = \infty$ since $[1, N_0] \subseteq R$. Let $\mathbb{P}_x^{\log}$ be the $1/n$-weighted measure on the odd $n \in (\sqrt{x}, x]$.

Proposition 10.1. $\psi_F(t) = \lim_{d\to\infty} \mathbb{P}_x^{\log}(\tau(n) > d \mid \mathrm{word}_2(n) = OO)$, a decreasing limit, and under $\mathrm{P}_\theta(C)$ — equivalently under any hypothesis of Sections 8–9 —

$$\psi_F(t) \;\leq\; 2\,(1+o(1))\Big(\frac{t}{2\log N_0}\Big)^{-(e(C)-\varepsilon)}.$$

Proof. The first claim is the definition of ψ_F together with $F \cap \{\text{odd}\} = \bigcap_d \{\tau > d\} \cap \{\text{odd}\}$. For the bound, cover $(\sqrt{x}, x]$ by dyadic blocks $(y, 2y]$; the OO-type odd starts of a block number $\frac{y}{4}(1 + O(y^{-\delta}))$ (the parity of $\lfloor n^{3/2} \rfloor$ over odd n is equidistributed with a power saving: a single exponential sum, the depth-two case of Paper B), the live starts at depth $d(y)$ number at most $\frac{y}{2}(\log 2y/\log N_0)^{-(e(C)-\varepsilon)}$ by Theorem 9.2, and $\{\tau > d\}$ decreases in d. The $1/n$-weighting only averages the block ratios. □

So the free term of the exact map is the infinite-depth live mass of the depth-two cylinder OO, and the hypotheses of Sections 8–9 bound it. There are not two frontier statements but one quantity — the live mass of odd-heavy parity words at depth $\asymp \log\log x$ — seen once as a limit (exact map) and once at finite depth with a rate (Tao-type bound).

10.2 Exact map = contagion + free term, and the critical exponent

The two halves of (6.1) are the two theorems of Sections 5 and 10. Dropping $\psi \geq 0$ gives the lower recursion $\varphi_F(t) \geq \frac{1}{2}\varphi_F(t/2) + \frac{1}{4}\varphi_F^{\mathrm{fib}}(3t/4)$, which in log-mass form $g = (t/2)\varphi$ is $g(t) \geq g(t/2) + \frac{1}{3}g^{\mathrm{fib}}(3t/4)$: the ideal depth-two contagion recursion of Section 5 with perfect fibers (the proved constants give $\lambda^{**} = 0.4480$, the ideal ones $\lambda_{\mathrm{ideal}} = 0.4927$). Dropping $\psi \leq 1$ instead gives the upper recursion for the failure density. Its homogeneous solutions $\varphi = t^{-e}$ satisfy $\frac{1}{2}2^e + c\,(4/3)^e = 1$ with $c = \frac{1}{4}$ for ideal fibers and $c = \frac{1}{3}$ with the two-sided pairing constant $\frac{2}{3}$ of Lemma 4.2:

$$e_{\mathrm{crit}} = 1 - \lambda_{\mathrm{ideal}} = 0.5073 \quad \text{(ideal)}, \qquad e_{\mathrm{crit}} = 0.3391 \quad \text{(pairing)}.$$

The rate threshold $e > 1 - \lambda^{**}$ of Theorem 7.2 (or $1 - \lambda^{***} = 0.4608$ with Appendix C) is therefore the homogeneous decay exponent of the exact map up to the fiber-constant gap: contagion's growth rate and the exact map's decay rate are the same number because they are the same recursion.

Corollary 10.2 (the exact map cannot replace contagion). Suppose $\psi_F(t) \leq t^{-e}$ with e as large as one likes. The upper recursion then gives $\varphi_F(t) \ll t^{-\min(e, e_{\mathrm{crit}})}$ and nothing more: with the proved constants $\varphi_F \ll t^{-0.339}$, ideally $\varphi_F \ll t^{-0.507}$, never $\varphi_F \equiv 0$. Theorem 1 gives $\varphi_F \gg t^{-(1-\lambda)}$ if $F \neq \emptyset$. The two bounds do not cross. What crosses Theorem 1 is the direct inclusion $F \cap (y, 2y] \subseteq \{\tau > d(y)\}$ applied to all starts — Theorem 7.2 — not the recursion. Conversely, if $F \neq \emptyset$ and ψ_F decayed faster than the critical rate, the sandwich would force $\varphi_F(t) \asymp t^{-0.507}$: a nonempty failure set with a small free term would have log-count exactly of the order $(\log x)^{0.4927}$.

10.3 The depth-uniformity budget

Section 9.3(e) says that a proof of the Tao-type bound must use information at depth $\to \infty$. The form of Theorem 8.3 says how uniform that information must be *along the cylinder-count route.*

Proposition 10.3 (depth-uniformity budget). Suppose a method establishes, for every depth d and every O-rooted word w of length d, $\#[w]_y = 2^{-(d-1)}\frac{y}{2} + O(y^{1-\delta_d})$ with $\delta_d = \delta_0 2^{-cd}$ for constants $\delta_0, c > 0$, and that this is its only error term. Then the bound of Theorem 8.3 follows from it at depth $d(y) = \lceil C \log_2 \log y \rceil$ if and only if $cC < 1$.

Proof. The bad words number at most 2^d, so the total error is $\le 2^d y^{1-\delta_d}$. This is $o(y(\log y)^{-e(C)})$ iff $\delta_d \log y - d \log 2 \ge (e(C) + o(1)) \log\log y$, i.e., with $2^d \asymp (\log y)^C$ and $\delta_d \asymp (\log y)^{-cC}$, iff $(\log y)^{1-cC} \gtrsim (C + e(C)) \log\log y$, which holds iff $cC < 1$. If $cC \ge 1$ the error term alone exceeds the fair main term. □

At $C = 19$ this requires the saving exponent to lose less than a factor $2^{1/19} = 1.037$ per depth; at $C = 41$, $2^{1/41} = 1.017$. Weyl differencing loses a factor 2^c with $c \ge 1$ per differencing (Paper B's chain $\frac{1}{24} \to \frac{1}{96}$ from depth three to four is $c = 2$), and with such a loss the cylinder-count route reaches depth $\approx \log_2 \log y$ — one unit of $L(y)$, enough for the all-E word to reach the floor, not for the Chernoff tail, which needs $C \ge 5$ units. The proposition is narrow by design: it concerns methods whose only available error estimate deteriorates exponentially in the depth and are used through the per-cylinder count of Theorem 8.3. A method that organizes its errors differently, that proves a polylogarithmic relative saving uniformly in the depth (as H(C, A) asks), or that addresses the pressure statement directly is not covered by it.

11. Numerical observations

Three numerical experiments accompany the theorems. They are exact integer computations with fixed random seeds (`research.juggler_sequence.fate_contagion`, `research.juggler_sequence.tao_reduction`; artifacts in Appendix B), they prove nothing, and they enter no proof. Detailed tables are kept in the repository notes [14].

Fibers, blocks, closure. Over $3375 \le m < 10^5$ and spot ranges at 10^6 and 10^7, the mean of G_m/H_m is 0.5000, 0.5004, 0.5027; the minimum on good fibers is 0.328 ($m = 1003635$, $\alpha_m \approx \frac{1}{3}$); no good fiber falls below $\frac{1}{3}H_m - 2$. On the blocks $m' \in [20, 60) \cup [200, 230) \cup [1000, 1010) \cup \{3000, 5000\}$ the deviation of $|U(m')|$ from its main term is at most $0.72\sqrt{|I(m')_{\mathrm{odd}}|}$, i.e. square-root scale. The closure of the Lean-verified seed $[1, 260]$ under the two productions of Section 3, computed exactly up to 10^9, has density between 0.45 and 0.49 on every dyadic block $(2^k, 2^{k+1}]$, $16 \le k \le 29$, against 0.25 for the E-only closure, and its realized recursion coefficients at 10^9 are 0.9998 for E (theory 1) and 0.3332 for OE (heuristic $\frac{1}{3}$); its OO-type share is 0 against a fair 0.30, the omitted production being exactly the free term of (6.1).

Survival above the certified floor. Because every $n \le N_0 = 3.5 \cdot 10^8$ reaches 1, the statistic "$J^t(n) \le N_0$ for some $t \le d$" is a finite computation, and its complement is the bad set of Theorem 8.3. For random odd starts in $(y, 2y]$, the fraction still above N_0 after d steps is compared with the odd-start fair-coin survival $\Pr[u_t > -L(y)\ \forall t \le d \mid u_1 = \log_2 3 - 1]$:

y	$L(y)$	$d = 10$	$d = 20$	$d = 30$	$d = 40$	ratio
10^{12}	0.526	0.250/0.250	0.082/0.084	0.043/0.043	0.0215/0.0219	0.98–1.01
10^{15}	0.841	0.291/0.309	0.100/0.106	0.050/0.055	0.026/0.028	0.91–0.95
10^{20}	1.249	0.360/0.383	0.148/0.153	0.068/0.068	0.038/0.038	0.94–1.02
10^{30}	1.826	0.442/0.440	0.207/0.206	0.107/0.103	0.053/0.051	1.00–1.07
10^{50}	2.558	0.618/0.617	0.267/0.264	0.149/0.143	0.087/0.084	1.00–1.05

Entries are empirical/fair (40000 starts at $10^{12}, 10^{15}, 10^{20}$, 20000 at $10^{30}, 10^{50}$); the last column is the range of empirical over fair for $d \ge 10$. On the same orbits the tilted odd share $s_\theta(t)$ of Section 9.2 lies in $[0.44, 0.56]$ for every $t \le 40$ and $\theta \in \{0.396, 0.6\}$, its cumulative excess over $\frac{1}{2}$ equals the positive-part noise of the effective tilted sample, and the live moment $\frac{1}{N}\sum_{\tau > d} e^{\theta o_d}$ is within 5–8% of its fair-coin value for $d \le 40$. These are observations about aggregates in the accessible range; the hypotheses of Section 9 concern $d(y) \to \infty$, which no sample can establish.

12. Limitations, and what a proof must do

What is proved. Theorems 1–5 and their corollaries, unconditionally in the sense that they depend only on the stated hypotheses and on the two exact productions. Appendix C depends on an additional analytic hypothesis.

What is not proved. Termination. No cycle is excluded, no divergent orbit is excluded, and no estimate in this paper or in Papers A and B controls the orbits of almost all starts down to a fixed bound. The numerical experiments are observations.

What a proof must do. Every argument available to us — cycle finance and the walk charge (Paper A), kernel equidistribution (Paper B), contagion (Section 5), the exact map (Section 6) — bounds a side of (6.1) that does not contain ψ_F. Sections 9–10 reduce the remaining problem to one statement:

> **The frontier statement.** The number of odd starts in $(y, 2y]$ whose orbit is still above the certified floor N_0 after $C \log_2 \log y$ steps is at most $y(\log y)^{-e}$ for some $e > 1 - \lambda^{**} = 0.552$ (0.461 under Appendix C).

By Theorem 3 this statement is equivalent to the conjecture. The rate threshold $1 - \lambda^{**}$ can be lowered by raising λ, and Section 5.7 prices that: the last rung reachable before K_3 is $\lambda \leq 0.7095$ with the present fiber sweep and ≤ 0.8414 with ideal fibers, i.e. $e > 0.2905$ and $e > 0.1586$, which move the least depth constant from $C = 18$ to $C = 14$ and $C = 11$. This lowers the frontier statement's constant; it does not remove its $\log\log y$ depth, and by Proposition 5.12 driving λ to 1 — which would make any positive rate suffice — needs O-runs of every length, hence K_3. Its weakest sufficient condition on the concentration route is the no-momentum hypothesis $\mathrm{M}_{\theta,q}(C)$: the tilted odd share of live starts exceeds $q < \log 2/\log 3$ by $o(\log\log y)$ in total over depths. That condition is one-sided, aggregated over cylinders, insensitive to any $o(\log\log y)$ initial depths and to towers biased below 0.836; it cannot be reached from cylinder statistics of bounded depth; and along the cylinder-count route it requires a saving that decays slower than $2^{-d/C}$ in the depth. The free term of the exact map is its OO-restriction at infinite depth, and contagion is the mechanism that turns it into the conjecture. What is needed, in the end, is a mean-over-cylinders statement about the parities of the nested floor powers $\lfloor \cdots \lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor \cdots \rfloor$ at depth of order $\log\log n$: the Terras step of the Juggler map.

Appendix A. Lean names

All in `formal/Problems/Juggler/FateContagion.lean` unless noted; the module compiles with `lake build Problems.Juggler` without `sorry`. Lean certifies the exact combinatorial identities listed here, not the analytic density estimates.

Statement	Lean
backward closure	`BackwardClosed`, `backwardClosed_iterate`
Lemma 2.1, fate classes closed	`reachesOne_backwardClosed`, `not_reachesOne_backwardClosed`, `ancestor_backwardClosed`, `escapes_backwardClosed`, `reachesOne_floorPower`, `escapes_floorPower`
Lemma 2.1, trichotomy	`Periodic`, `periodic_of_repeat`, `fate_trichotomy` (from `cycles_or_escapes`)
Lemma 2.1, exclusion	`reachesOne_bounded`, `reachesOne_not_escapes`, `cycle_basin_bounded`, `cycle_basin_not_escapes`, `reachesOne_not_cycle_basin`, `periodic_iterate_mod`

Statement	Lean
Lemma 3.1 (even block)	`floorPower_even_block`, `even_block_mem`, `even_block_card`
Lemma 3.2 (*OE* fiber)	`sqrt_sqrt_eq_iff`, `floorPower_oe_fiber`, `oe_fiber_mem`, `oe_fiber_disjoint`
Theorem 6.1 (odd generation)	`ForwardClosed`, `reachesOne_forwardClosed`, `not_reachesOne_forwardClosed`, `escapes_forwardClosed`, `mem_iff_floorPower_mem`, `exists_odd_ancestor`, `exists_odd_ancestor_ge_three`, `nonempty_iff_odd_image_mem`, `odd_mem_iff`
Lemma 8.1 (envelope descent)	`iterate_le_of_envelope`, `mem_of_envelope_floor`, `reachesOne_of_itinerary_envelope`; power envelope `power_bound_word` (Paper A layer)
Lean floor $N_0 = 260$	`reachesOne_of_lt_two_hundred_sixty_one`
Lemmas 4.1, 4.1', 4.2–4.3, Proposition 4.4 ($C_0 = 250$), Lemma 5.1, Theorem 5.3, Sections 7–10, Appendix C	human proofs

Appendix B. Constants and artifacts

Exponents. Roots of $2^{-\lambda} + \sum_i c_i e_i^\lambda = 1$:

recursion	terms (c_i, e_i)	root
elementary sweep only	$(\frac{2}{21}, \frac{3}{4})$	0.1385
block average only (λ^*)	$(\frac{1}{3}, \frac{3}{8})$	0.3774
block average + sweep (adversarial 1/7)	$(\frac{5}{21}, \frac{3}{8}), (\frac{2}{21}, \frac{3}{4})$	0.4051
block average + pairing (λ^{**}, Theorem 1)	$(\frac{1}{9}, \frac{3}{8}), (\frac{2}{9}, \frac{3}{4})$	0.4480
+ *OOEEE* on even blocks (λ^{***}, Appendix C, conditional)	$\dots, (\frac{1}{9}, \frac{9}{32})$	0.5392
depth-two ideal	$(\frac{1}{3}, \frac{3}{4})$	0.4927
+ *OOOEE*, *OOEOE* (closed: fibers $P^{5/32}$; Lemma 3.9 leftover $P^{89/96}$)	$\dots, (\frac{2}{27}, \frac{27}{64})$	0.5561
+*OEOEE* (elementary, constants pending; Section 5.7)	$\dots, (\frac{1}{27}, \frac{9}{32})$	0.4801
+*OEOEE* and *OOEEE*	$\dots, (\frac{4}{27}, \frac{9}{32})$	0.5665
+ the whole family $V_k = (OE)^{k-1}OEE$	$\dots, (3^{-(k+2)}, (\frac{3}{4})^k \frac{3}{8})_{k \ge 1}$	0.4927
+ that family and *OOEEE*		0.5769
O-runs ≤ 2 controlled, present sweep (Section 5.7)	transfer matrix	0.6247
O-runs ≤ 2, ideal fibers	transfer matrix	0.7180
O-runs ≤ 3 (last rung before K_3), present sweep	transfer matrix	0.7095
O-runs ≤ 3, ideal fibers	transfer matrix	0.8414
ceiling, all *O*-runs at the ideal share (Proposition 5.12)	$2^{-\lambda} + \frac{1}{3}(\frac{3}{2})^\lambda = 1$	1

Depth constants. Chernoff exponent $e(C)$: 0.480 (18), 0.527 (19), 0.574 (20), 0.621 (21), 0.812 (25), 1.054 (30). Least C for the rate threshold $1-\lambda^{**} = 0.5520$ / conditional $1-\lambda^{***} = 0.4608$: fair Chernoff 20/18; one-sided Azuma at $q = 0.5, 0.55, 0.60, 0.62$: $20, 44, 240, 1715$ / $18, 39, 206, 1451$; pressure (biased Chernoff) at the same q: $20, 43, 230, 1618$ / $18, 38, 198, 1369$. Depth $d(y) = \lceil 20L(y)\rceil$ with $N_0 = 3.5\cdot 10^8$: 25 at 10^{20}, 72 at 10^{100}, 138 at 10^{1000}, 204 at 10^{10000}.

Artifacts (repository `sneakyweasel/balanced_ternary`; SHA-256):

file	hash
`data/research/juggler/fate_contagion/summary.json`	`85030bcb5f4964b814b101683c2721efa5f7299b687afa9e399febe603`
`data/research/juggler/tao_reduction/summary.json`	`76c0ae713d34569cdf8efd90231712f9281f7d5083346a2d9384c536f7`
`formal/Problems/Juggler/FateContagion.lean`	`cac6a00884346fcc98a03603bf919e03a681f8f66b8b55f3cff9d336e8`
`src/research/juggler_sequence/fate_contagion.py`	`34f8cff465e00187cb85e1dc9a75a3b250caa4c8f38a3bc41aef29f68b`
`src/research/juggler_sequence/tao_reduction.py`	`90f930bd604f6aa38c3a5ec8265d270d218240cdb358b20b19cad98dc4`
`docs/theory/figures/render_paper_c_figures.py`	`5e434450835aadb4ed5ed2cbed00cc33f774996ba9ef229cb0434fc677`

The numerical experiments of Section 11 use fixed random seeds and are reproduced by `python -m research.juggler_sequence.fate_contagion` and `python -m research.juggler_sequence.tao_reduction`; the figures by `python docs/theory/figures/render_paper_c_figures.py`.

Appendix C. A conditional strengthening

This appendix depends on one analytic hypothesis beyond the main text. No statement of Sections 1–12 uses it; it improves the constants of Theorems 1, 3 and 4 and nothing else.

C.1 The hypothesis

For odd n put $m = \lfloor n^{3/2}\rfloor$, $v = \lfloor m^{3/2}\rfloor$, $\psi(y) = (-1)^{\lfloor y\rfloor}$, and

$$\psi_1 = \psi(n^{3/2}), \qquad \psi_2 = \psi(m^{3/2}), \qquad \psi_3 = \psi(v^{1/2}),$$

the signs of the parities of m, of $\lfloor m^{3/2}\rfloor$ and of $\lfloor v^{1/2}\rfloor$ — the second, third and fourth letters of the itinerary of n along the branch OOE.

Hypothesis L (localized triple parity discrepancy with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \ge P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{1/2}$, every $(a,b,c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$, and every integer ℓ with $|\ell| \le P^{1/24}$,

$$\Big|\sum_{n\in I,\ n \text{ odd}} \psi_1^a \psi_2^b \psi_3^c\, e\big(\tfrac{\ell}{2} n^{9/16}\big)\Big| \;\le\; Y\, P^{-1/24+\varepsilon}.$$

Status. Hypothesis L is Theorem 4.12 of the working draft [12] (Section 4.5 there), whose Lemma 4.10 removes the twist by partial summation after the Weyl differencing and whose proof runs the seven-step argument of its Theorems 4.4 and 4.7 (exact linearization of the nested floors, differencing with $H = P^{1/12}$, a cell decomposition, second-derivative tests per cell) over a sub-dyadic interval, exhibiting the three absolute terms — two partial end cells $5.2P^{3/8}$, the pure passenger $3.8P^{7/16}$, the majorant end cells $17P^{1/4}$ — that do not scale with the number of summands. Because [12] is an unrefereed working draft, we keep the statement as a hypothesis here and rely on nothing but the statement; what *is* proved in this appendix is everything that follows from it. (The derivation below coincides with Corollary 4.13 of [12].)

C.2 The $OOEEE$ production on even blocks

For $m' \ge 2$ let $I(m') = [m'^{32/9}, (m'+1)^{32/9})$, an interval of length $Y = \frac{32}{9} m'^{23/9}(1 + O(1/m'))$ at scale $P = m'^{32/9}$, so that $Y \asymp P^{23/32} \ge P^{1/2}$. Let

$$\mathcal{O}(m') = \{n \text{ odd} \in I(m') :\ \mathrm{word}_5(n) = OOEEE,\ J^5(n) = m'\}.$$

Lemma C.1 (transparent nesting at the fifth letter). For odd $n \geq 3$, with m, v as above, $0 \leq n^{9/16} - v^{1/4} \leq n^{-15/16}$. Consequently $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$ unless $\{n^{9/16}\} < n^{-15/16}$.

Proof. $v \leq m^{3/2} \leq n^{9/4}$ gives $v^{1/4} \leq n^{9/16}$. For the other direction, $m \geq n^{3/2} - 1$ and $v \geq m^{3/2} - 1$ give $v \geq (n^{3/2}-1)^{3/2} - 1 \geq n^{9/4} - \frac{3}{2}n^{3/4} - 1$, and then, by the mean value theorem for $u \mapsto u^{1/4}$ on $[v, n^{9/4}]$, $n^{9/16} - v^{1/4} \leq (n^{9/4} - v)/(4v^{3/4}) \leq (\frac{3}{2}n^{3/4}+1)/(4v^{3/4})$. Since $v \geq \frac{1}{2}n^{9/4}$ for $n \geq 3$, $v^{3/4} \geq 2^{-3/4}n^{27/16}$, and the bound is at most $2^{3/4}(\frac{3}{2}n^{3/4}+1)/(4n^{27/16}) \leq n^{-15/16}$ for $n \geq 3$. The floors of two numbers in $[n^{9/16} - n^{-15/16}, n^{9/16}]$ differ only if an integer lies in that interval, i.e. if $\{n^{9/16}\} < n^{-15/16}$. □

Lemma C.2 (the exceptional set is small). Let $\mathcal{E}(I) = \{n \text{ odd} \in I : \{n^{9/16}\} < n^{-15/16}\}$. For $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$, $|\mathcal{E}(I)| \ll YP^{-1/16}$.

Proof. $\mathcal{E}(I) \subseteq \{n \text{ odd} \in I : \{n^{9/16}\} \in [0, P^{-15/16})\}$. By the Erdős–Turán inequality [15, Theorem 2.5], for any $H \geq 1$,

$$|\mathcal{E}(I)| \;\leq\; \tfrac{Y}{2}P^{-15/16} + C\Big(\frac{Y}{H} + \sum_{h=1}^{H}\frac{1}{h}\Big|\sum_{n \in I \text{ odd}} e(hn^{9/16})\Big|\Big).$$

Write $n = 2k+1$ and $F_h(k) = h(2k+1)^{9/16}$; then $F_h'(k) = \frac{9}{8}h(2k+1)^{-7/16}$ is monotone with $0 < F_h' \leq \frac{9}{8}hP^{-7/16} \leq \frac{1}{2}$ for $h \leq H := \frac{4}{9}P^{7/16}$, so the Kusmin–Landau inequality [13, Theorem 2.1] gives $|\sum_k e(F_h(k))| \ll 1/\min F_h' \ll P^{7/16}/h$. Hence $|\mathcal{E}(I)| \ll YP^{-15/16} + YP^{-7/16} + P^{7/16}$, and $P^{7/16} \leq YP^{-1/16}$ for $Y \geq P^{1/2}$. □

Lemma C.3 (a Fourier expansion of the fifth letter). Let $J = \lfloor P^{1/24} \rfloor$. There are coefficients b_q, $0 < |q| \leq J$, with $|b_q| \ll 1/|q|$, and a nonnegative trigonometric polynomial Δ_J of degree J with all coefficients at most $1/(J+1)$, such that for every real y, $\big|\psi(y) - \sum_{0<|q|\leq J} b_q e(qy/2)\big| \leq 2\Delta_J(y/2)$. Moreover, for $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$,

$$\sum_{n \in I \text{ odd}} \Delta_J(\tfrac{1}{2}n^{9/16}) \ll YP^{-1/24} + P^{7/16}\log P, \qquad \Big|\sum_{n \in I \text{ odd}} e(\tfrac{q}{2}n^{9/16})\Big| \ll \frac{P^{7/16}}{|q|} \quad (0 < |q| \leq J).$$

Proof. $\psi(y) = 2\cdot\mathbf{1}[\{y/2\} < \frac{1}{2}] - 1$, and Vaaler's theorem [10] approximates the indicator of $[0, \frac{1}{2})$ modulo 1 by a trigonometric polynomial of degree J with coefficients $\ll 1/|q|$ and constant term $\frac{1}{2}$, with error at most a majorant Δ_J of the stated form; the constant terms cancel in $2\cdot\mathbf{1} - 1$. For the two sums, write $n = 2k+1$: the phase $\frac{q}{2}(2k+1)^{9/16}$ has derivative $\frac{9q}{16}(2k+1)^{-7/16}$, monotone, of absolute value between $\frac{9}{16}|q|(2P)^{-7/16}$ and $\frac{9}{16}P^{1/24-7/16} < \frac{1}{2}$, so Kusmin–Landau gives $\ll P^{7/16}/|q|$. The Δ_J sum is its constant term, $\leq Y/(J+1)$, plus $(J+1)^{-1}\sum_{0<|r|\leq J}|\sum_n e(\frac{r}{2}n^{9/16})| \ll P^{7/16}\log J/J$. □

Proposition C.4 (conditional on Hypothesis L). For $m' \geq 2$,

$$|\mathcal{O}(m')| = \tfrac{1}{16}\#\{n \text{ odd} \in I(m')\} + O_\varepsilon\big(|I(m')|\, m'^{-4/27+\varepsilon}\big),$$

every $n \in \mathcal{O}(m')$ has $J^4(n) \in E(m')$, and for a backward-closed A and $m' \in A$, $\mathcal{O}(m') \subseteq A$ with log-mass at least $\frac{1}{9m'}(1 - O(m'^{-4/27+\varepsilon}))$.

Proof. Along $OOEEE$ the states $J^0(n), \dots, J^4(n)$ are n (odd), m (odd), v (even), $\lfloor v^{1/2} \rfloor$ (even), $\lfloor\sqrt{\lfloor v^{1/2}\rfloor}\rfloor = \lfloor v^{1/4} \rfloor$ (even; the inner identity is the exact nesting of square roots, Lemma 3.2), and $J^5(n) = \lfloor\sqrt{J^4(n)}\rfloor$. If $J^5(n) = m'$ then $J^4(n) \in [m'^2, (m'+1)^2)$ is even, i.e. $J^4(n) \in E(m')$, so $n \in A$ by Lemma 3.1 and backward closure. For the count, let $I = I(m')$, $P = m'^{32/9}$, $Y = |I|$. For $n \in I \setminus \mathcal{E}(I)$, Lemma C.1 gives $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$, and $n \in I$ means $n^{9/16} \in [m'^2, (m'+1)^2)$; hence for such n with $\text{word}_5(n) = OOEEE$, $J^4(n) = \lfloor n^{9/16} \rfloor \in [m'^2, (m'+1)^2)$ and $J^5(n) = m'$ automatically. Therefore

$$\big|\,|\mathcal{O}(m')| - \#\{n \text{ odd} \in I : \text{word}_5(n) = OOEEE\}\,\big| \leq |\mathcal{E}(I)| \ll YP^{-1/16}$$

by Lemma C.2. Next, for odd n the indicator of $OOEEE$ is $\frac{1}{16}(1-\psi_1)(1+\psi_2)(1+\psi_3)(1+\psi_4)$ with $\psi_4 = \psi(v^{1/4})$: the factor $1-\psi_1$ enforces m odd, so that $J^2(n) = v$ and ψ_2 is its parity sign; $1+\psi_2$ enforces v

even, so that $J^3(n) = \lfloor v^{1/2} \rfloor$ and ψ_3 is its sign; $1+\psi_3$ enforces $J^3(n)$ even, so that $J^4(n) = \lfloor v^{1/4} \rfloor$ and ψ_4 is its sign. Expanding the product gives the main term $\frac{1}{16}\#\{n \text{ odd} \in I\}$ and fifteen sign sums $\pm\frac{1}{16}\sum_n \psi_1^a\psi_2^b\psi_3^c\psi_4^d$, $(a,b,c,d) \neq 0$.

Sums with $d = 0$. These are Hypothesis L with $\ell = 0$: each is $\leq YP^{-1/24+\varepsilon}$.

Sums with $d = 1$. Replace $\psi_4 = \psi(v^{1/4})$ by $\psi(n^{9/16})$: by Lemma C.1 the two agree off $\mathcal{E}(I)$, so the change is $\leq 2|\mathcal{E}(I)| \ll YP^{-1/16}$. Then expand $\psi(n^{9/16})$ by Lemma C.3:

$$\sum_n \psi_1^a\psi_2^b\psi_3^c\,\psi(n^{9/16}) = \sum_{0<|q|\leq J} b_q \sum_n \psi_1^a\psi_2^b\psi_3^c\, e\big(\tfrac{q}{2}n^{9/16}\big) + O\Big(\sum_n \Delta_J\big(\tfrac{1}{2}n^{9/16}\big)\Big),$$

using $|\psi_i| = 1$ for the error. If $(a,b,c) \neq 0$, each inner sum is Hypothesis L with $\ell = q$, $|q| \leq J \leq P^{1/24}$, and $\sum_q |b_q| \ll \log P$, so the total is $\ll YP^{-1/24+\varepsilon}\log P$. If $(a,b,c) = 0$ the inner sums are the pure sums of Lemma C.3, $\ll P^{7/16}/|q|$, and their total against $|b_q| \ll 1/|q|$ is $\ll P^{7/16}$. The majorant term is $\ll YP^{-1/24} + P^{7/16}\log P$ by Lemma C.3.

Collecting, with $P^{7/16}\log P \leq YP^{-1/16}\log P$ for $Y \geq P^{1/2}$, every error is $\ll YP^{-1/24+\varepsilon}$, and $P^{-1/24} = m'^{-4/27}$. For the log-mass, each $n \in \mathcal{O}(m')$ has $1/n \geq (m'+1)^{-32/9}$, and $\frac{1}{16}\cdot\frac{1}{2}\cdot\frac{32}{9}m'^{23/9} = \frac{1}{9}m'^{23/9}$. $\square$

C.3 The improved exponent

Theorem C.5 (conditional on Hypothesis L). Theorem 5.3 holds for every $\lambda < \lambda^{***} = 0.5392\ldots$, the root of

$$2^{-\lambda} + \tfrac{1}{9}\big(\tfrac{3}{8}\big)^\lambda + \tfrac{2}{9}\big(\tfrac{3}{4}\big)^\lambda + \tfrac{1}{9}\big(\tfrac{9}{32}\big)^\lambda = 1.$$

Proof. Add to the three families of Section 5.1 a fourth: for $m' \in A \cap (x^{9/64}, x^{9/32} - 1]$, the set $\mathcal{O}(m')$; then $I(m') \subseteq (\sqrt{x}, x]$ because $m'^{32/9} > \sqrt{x}$ and $(m'+1)^{32/9} \leq x$. Its members are odd with odd $\lfloor n^{3/2} \rfloor$, hence disjoint from the even members and from the *OE*-type members of the first three families, and distinct m' give disjoint $I(m')$. By Proposition C.4 the family has log-mass at least $\frac{1}{9}(1-\varepsilon_7(x))(g_A(9t/32) - 2x^{-9/32})$ with $\varepsilon_7 \to 0$, so (5.2) gains the term $(\frac{1}{9} - \varepsilon_7)g_A(9t/32)$. Lemma 5.1 applies with the additional pair $(e_4, c_4) = (\frac{9}{32}, \frac{1}{9})$ and the new root. $\square$

Downstream, every constant of Sections 7–10 improves: the rate threshold of Theorems 7.2–7.3 becomes $e > 1 - \lambda^{***} = 0.4608$; the least depth constant of Corollary 8.4 and Theorem 9.2 is $C = 18$ ($e(18) = 0.480$); in the one-sided form $C(0.5) = 18$, $C(0.55) = 39$. Pairing plus *OOEEE* sits at 0.5392, above the depth-two ceiling 0.4927, because *OOEEE* is an extra production. A further production from the words *OOOEE* and *OOEOE* on even blocks would give $\lambda = 0.5561$ and $C = 18$; those fibers have length $P^{5/32}$, below the threshold of the localized triple, and Lemma 3.9's trivial bound on the kernel proof is the whole interval, so the analogue of Hypothesis L for the kernel theorem of [12] is not available (fate-contagion note §7.4). Hypothesis L concerns consecutive odd starts in an interval and says nothing about the odd images S_{odd}; the free term ψ_F is untouched by this appendix.

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The exact combinatorial layer was developed with machine assistance and checked in Lean 4; the analytic counting, the reduction, and the prose are the author's responsibility. The repository notes from which this paper is distilled [14] record every intermediate step, including the ideas that failed.

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